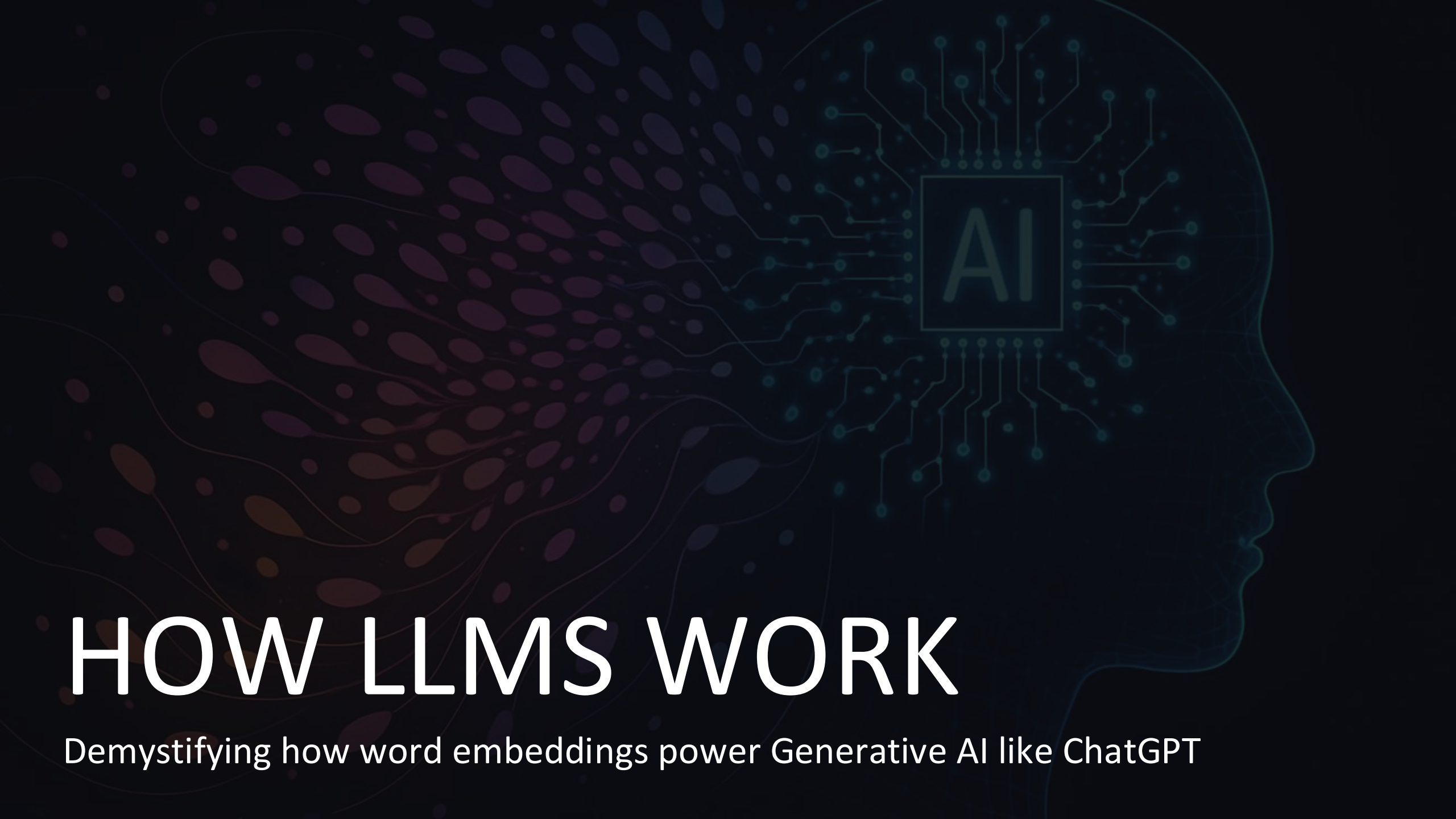




HOW LLMS WORK

Demystifying how word embeddings power Generative AI like ChatGPT

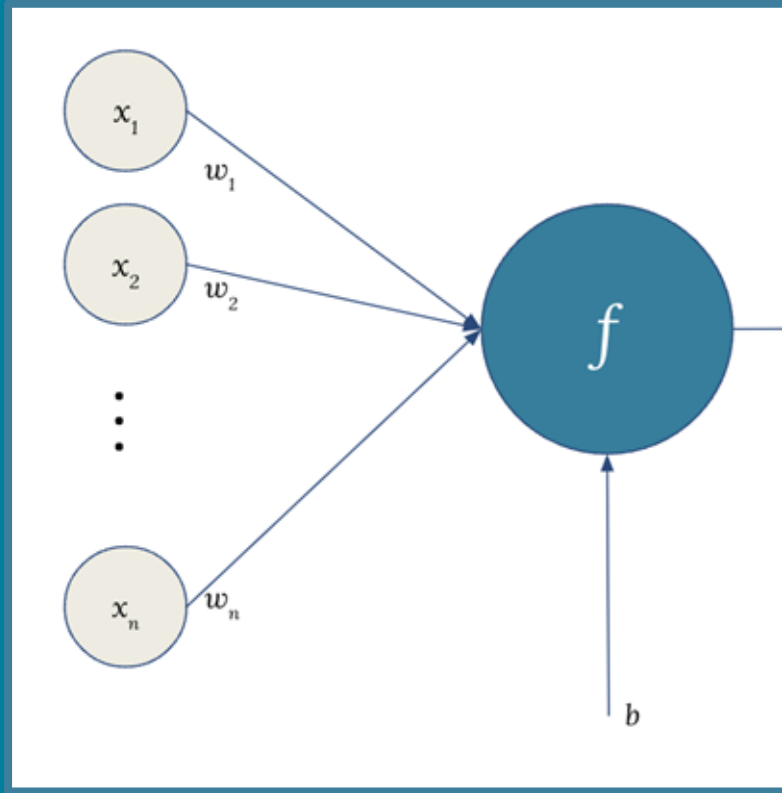
A little about me

I am interested in how machine learning methods can be leveraged to effect positive change in the world, and especially interested in ensuring that those methods are fair and unbiased.

Currently: Researching the integration of state-of-the-art large language models into the New York Times' reporting and products, and teaching Intro to AI at The Coding School

Previously: Columbia University and Google

Schedule



Segment 1 Introduction to LLMs and Generative AI

Segment 2: Word Embeddings

- What are word embeddings?
- How are embeddings formed?
- How can we do math with human language?

Break (5-10 minutes)

Segment 3: Word Embeddings in Generative Language Models

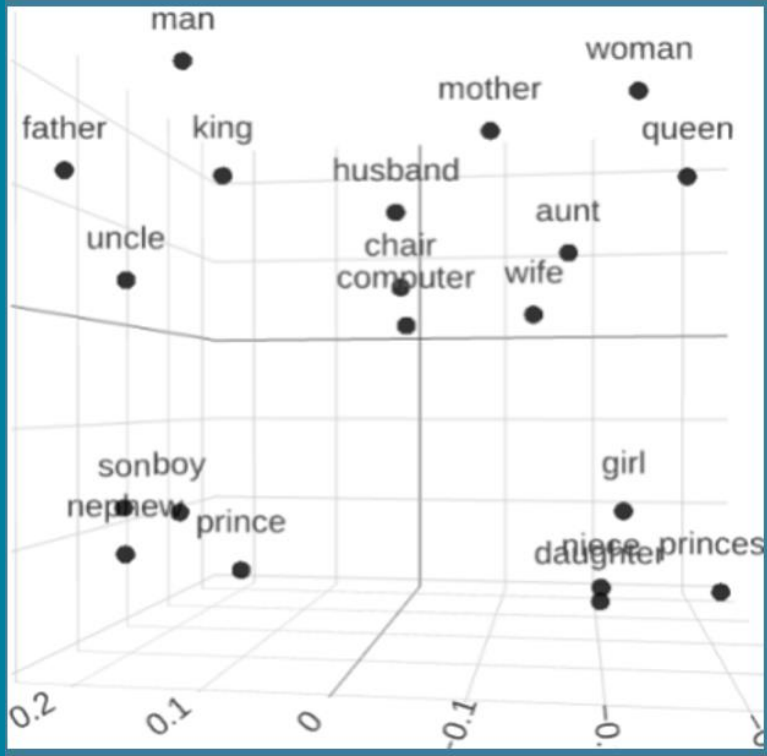
- How are word embeddings used in generative language models?
- What works well (and what can go wrong) when we use word embeddings?

Segment 4: Other Use Cases for Embeddings

- Summarization
- Clustering and Recommendation
- Retrieval Augmented Generation
- Multi-modal systems

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Schedule

Complete
this
sentence...

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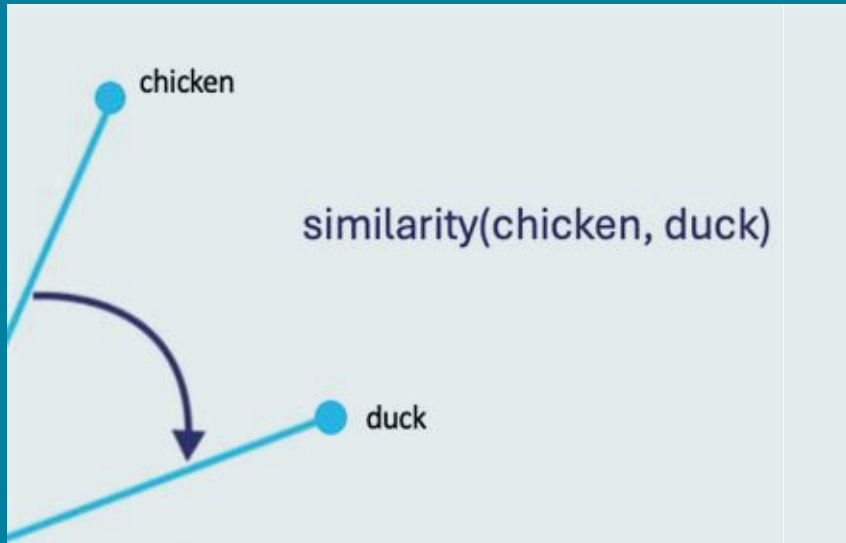
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Audience Poll Question

Have you used a generative LLM / chatbot before? (multiple choice)

- OpenAI's ChatGPT
- Google's Gemini
- Microsoft's Copilot
- Anthropic's Claude
- Other
- Never used a generative AI chatbot

Audience Poll Question

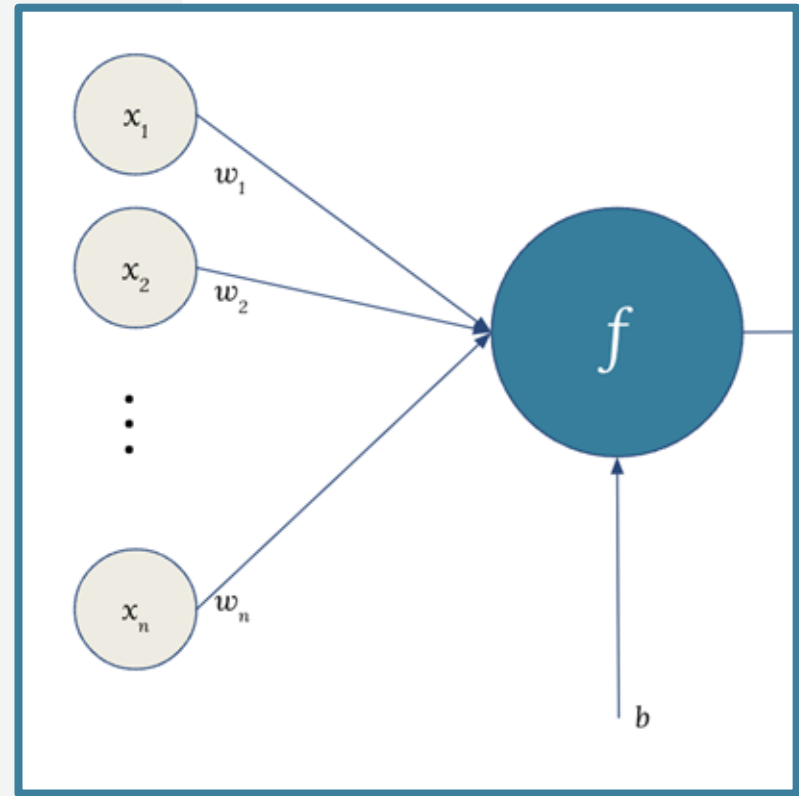
Which of the following have you heard of or used before?
(multiple choice)

- GPT-4
- Huggingface
- BERT
- DeepSeek
- LLaMA
- Neural networks
- Transformer architecture
- Word Embeddings

Section 1: Introduction to LLMs and Generative AI

By the end of this section you will:

- Understand what the term LLM means
- Understand what the term Generative AI means
- Have some insight into how Artificial Intelligence / Machine Learning works in general



Terms

- Artificial Intelligence (AI) / Machine Learning (ML)
- Model / Algorithm

LLM *Large Language Model*

A model that is trained on a large amount of text data and can **understand** human text and/or **generate** human-like text



ChatGPT



ChatGPT





ChatGPT

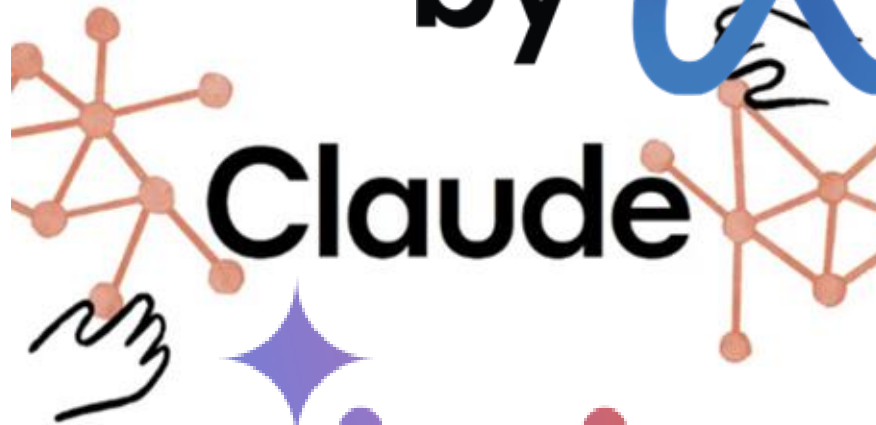


Gemini



ChatGPT

LLaMA
by  Meta

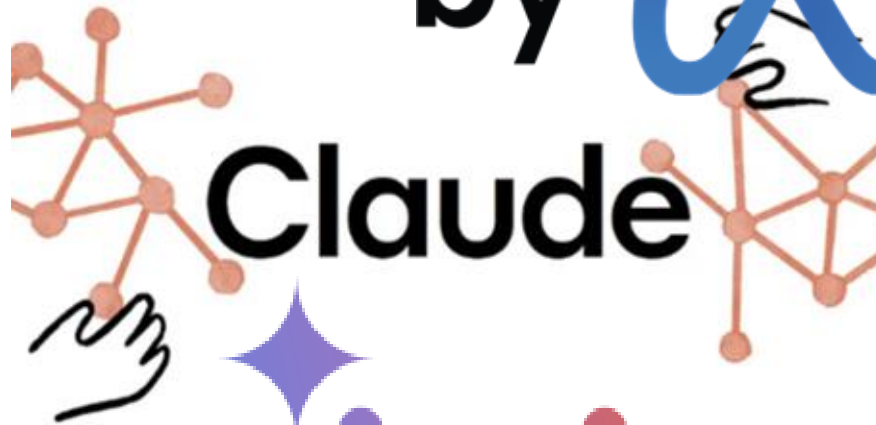


Gemini



ChatGPT

LLaMA
by  Meta



Gemini





ChatGPT

LLaMA
by  Meta



deepseek

Gemini



Generative AI

A **generative** model is one that can create new content based on data it's been trained on

Generative AI

A **generative** model is one that can create new content based on data it's been trained on

- **Image** generation
- **Video** generation
- **Music** generation
- **Text** generation



Generative LLM



You

write a haiku about cleaning the bathroom



ChatGPT

Sinks gleam, tiles so bright,
Scrub away grime's stubborn fight,
Freshness fills the light.



Generative LLM

 **You**

write a haiku about cleaning

 **ChatGPT**

Sinks gleam, tiles so bright
Scrub away grime's stubborn
Freshness fills the light.



 **You**

give me some ideas of the best exercises to do with a bad back

 **ChatGPT**

Exercising with a bad back requires caution and focus on low-impact activities that strengthen the core muscles and improve flexibility without exacerbating existing pain. Here are some ideas:

1. **Walking:** A simple yet effective low-impact exercise that promotes cardiovascular health and helps maintain mobility without straining the back.
2. **Swimming or Water Aerobics:** These activities provide excellent cardiovascular benefits and strengthen muscles without stressing the spine. The buoyancy of water reduces the impact on the back.
3. **Cycling:** Either stationary or outdoor biking can be beneficial for people with back issues. Ensure proper bike fit and maintain a comfortable, upright posture while cycling.
4. **Pilates:** Pilates focuses on core strength, flexibility, and stability, which can help alleviate back pain by improving posture and supporting the spine.
5. **Yoga:** Gentle yoga poses can improve flexibility, strengthen core muscles, and promote

Generative LLM

 **You**
write a haiku about clean

 **ChatGPT**

Sinks

Scrub

Fresh



 **You**
give me some ideas of the best exercises to do with a bad back

 **ChatGPT**
Exercising with a bad back requires caution and focus on low-impact activities that strengthen the core muscles and improve flexibility without exacerbating existing pain. Here are some ideas:

 **You**
what started world war I?

 **ChatGPT**
World War I began primarily due to a complex web of political alliances, militarism, imperialism, and nationalism that had been building up in Europe for decades before the war erupted in 1914. The immediate trigger was the assassination of Archduke Franz Ferdinand of Austria-Hungary by a Bosnian Serb nationalist in Sarajevo on June 28, 1914. This event ignited tensions between various European powers, leading to a chain reaction of declarations of war. Austria-Hungary declared war on Serbia, leading to Russia's mobilization to support Serbia. Germany, aligned with Austria-Hungary, declared war on Russia and its ally France. The war escalated further with other countries joining in, eventually engulfing much of the world in conflict.



notes cardiovascular health and

lent cardiovascular benefits and
cy of water reduces the impact on

for people with back issues. Ensure
while cycling.

ility, which can help alleviate back

core muscles, and promote

Generative LLM

You
write a haiku about sleep



ChatGPT

Sinks

Scrub

Fresh



You
give me some ideas of the best exercises to do with a bad back

You
write a short story of 100 words about a bird that cannot find love



ChatGPT

In the heart of the forest lived a bird with feathers of the purest white. Despite its elegance, love eluded it. Day after day, it watched as pairs of birds fluttered by, their melodies weaving tales of blissful companionship. Yet, for this solitary bird, the sky remained empty. It sang its song, hoping to attract a mate, but its notes echoed unanswered through the trees. Seasons changed, but its loneliness endured. Until one evening, as the sun dipped below the horizon, a bird with plumage of vibrant hues landed nearby. With a tender trill, their melodies intertwined, and love finally found its way home.

various European powers, leading to a chain reaction of declarations of war. Austria-Hungary declared war on Serbia, leading to Russia's mobilization to support Serbia. Germany, aligned with Austria-Hungary, declared war on Russia and its ally France. The war escalated further with other countries joining in, eventually engulfing much of the world in conflict.



But how does this **work**?

How can an algorithm learn to **interpret** and **produce** human language?

Training

Developing the model

Input



sianc/Shutterstock
MDavidova / Shutterstock

Model

Output

“cat” / “dog”

Training

Developing the model

Input



Model



Output

Training

Developing the model

Input



Model



Output

“dog”?

Training

Developing the model

Input



Model



Output

“dog”?



Training

Developing the model

Input



Model



Output

“dog”?



Training

Developing the model

Input



MDavidova / Shutterstock



Model



Output

“cat”?



Training

Developing the model

Input



sianc/Shutterstock



Model



Output

“cat”?



Training

Developing the model

Input



sianc/Shutterstock



Model



Output

“cat”?



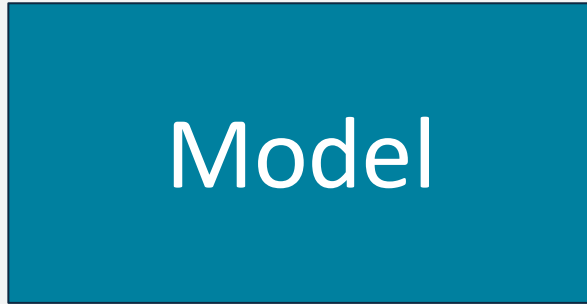
Training

Developing the model

Input



Model



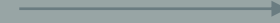
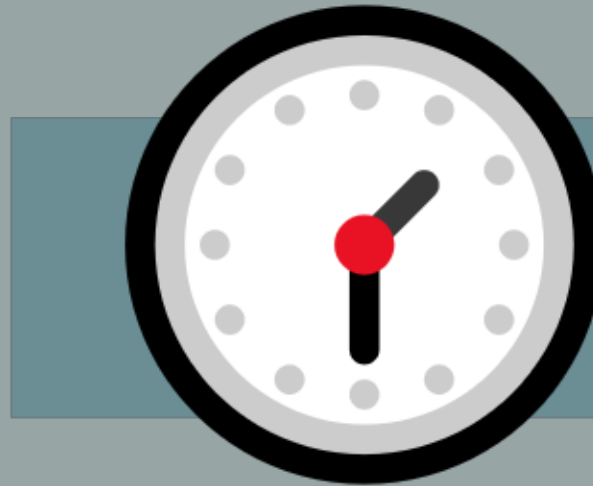
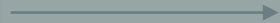
Output

“cat”?



Training

Input



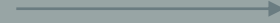
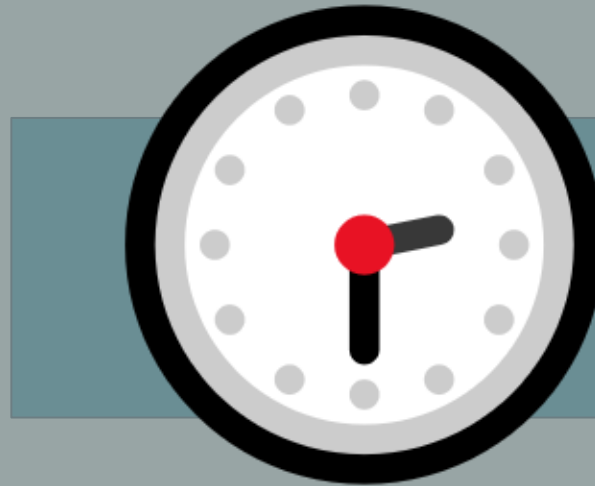
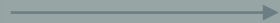
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Training

Input



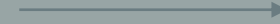
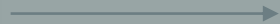
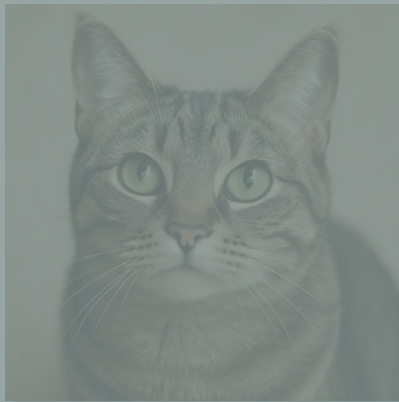
Output

“cat”?



Training

Input



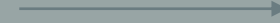
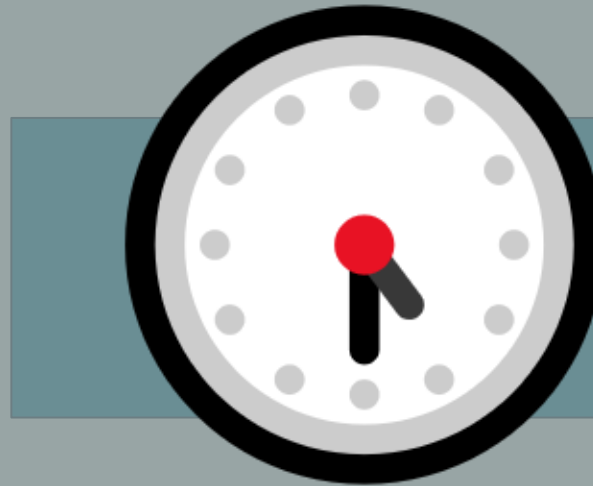
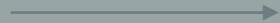
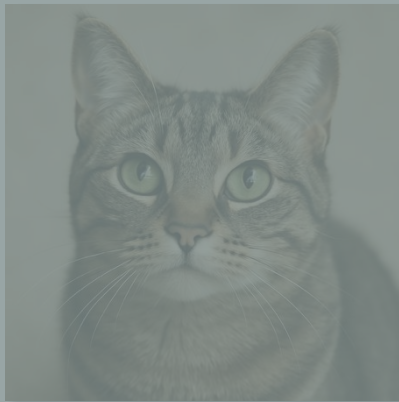
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Training

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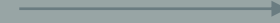
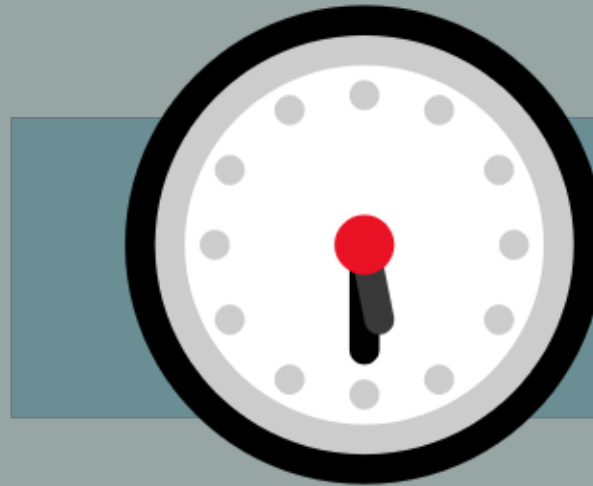
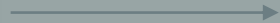
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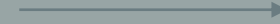
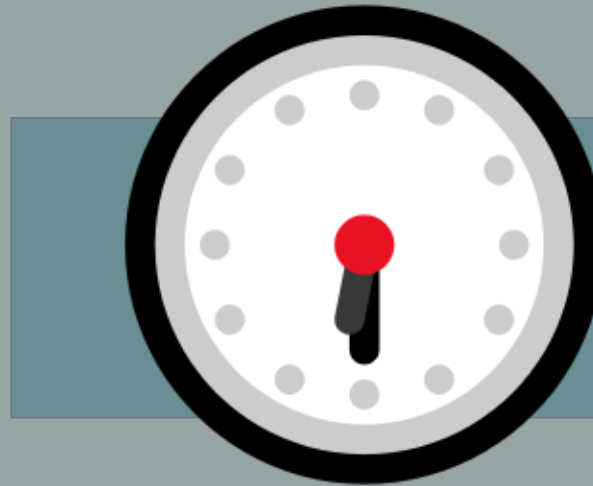
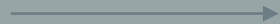
Output

“cat”?



Training

Input



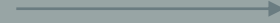
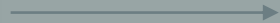
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Training

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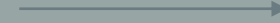
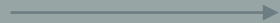
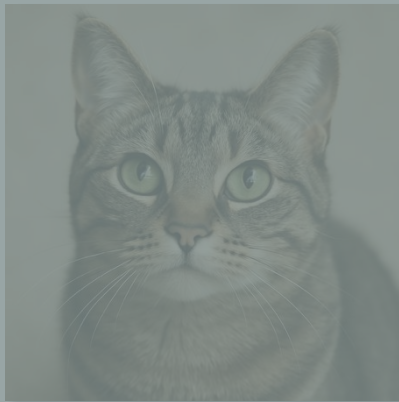
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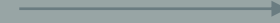
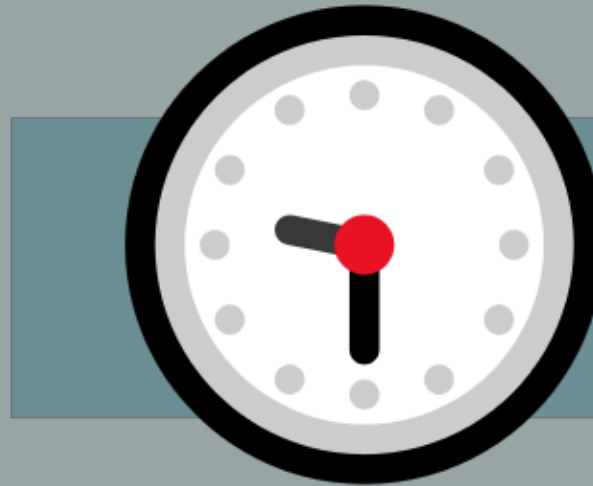
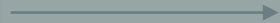
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Training

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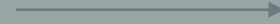
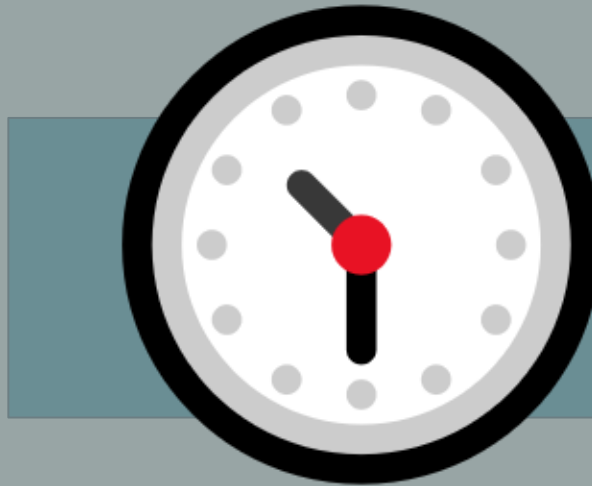
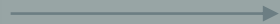
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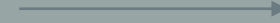
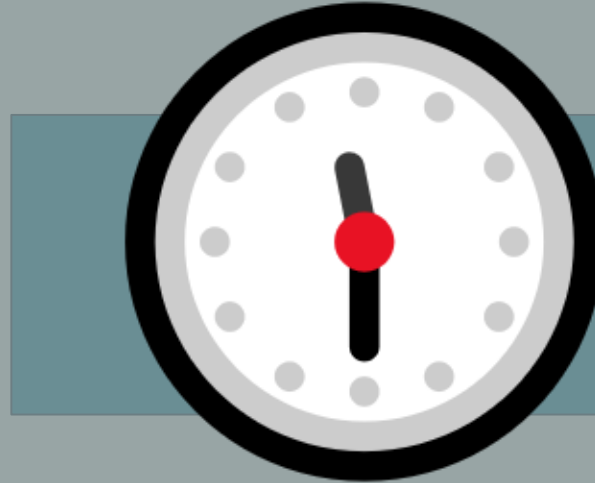
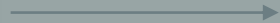
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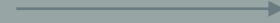
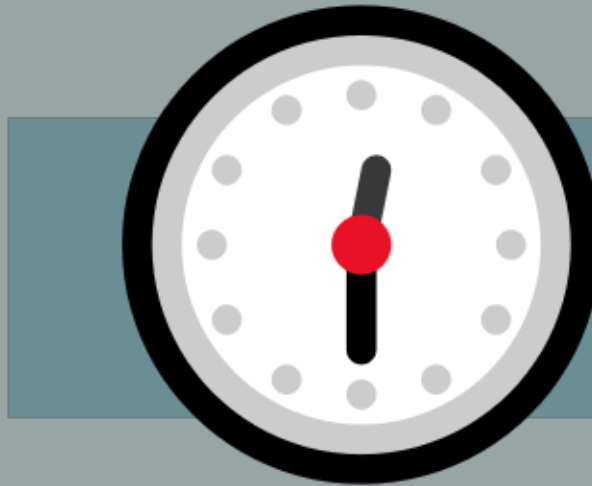
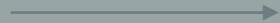
Output

“cat”?



Training

Input



Output

“cat”?



Inference

Once the model is out in the world

Input



Model



Output

“cat”

Inference

Once the model is out in the world

Input



Model



Output

“cat”

?

Inference

Once the model is out in the world

Input



Model



Output

Inference

Once the model is out in the world

Input



Model



Output

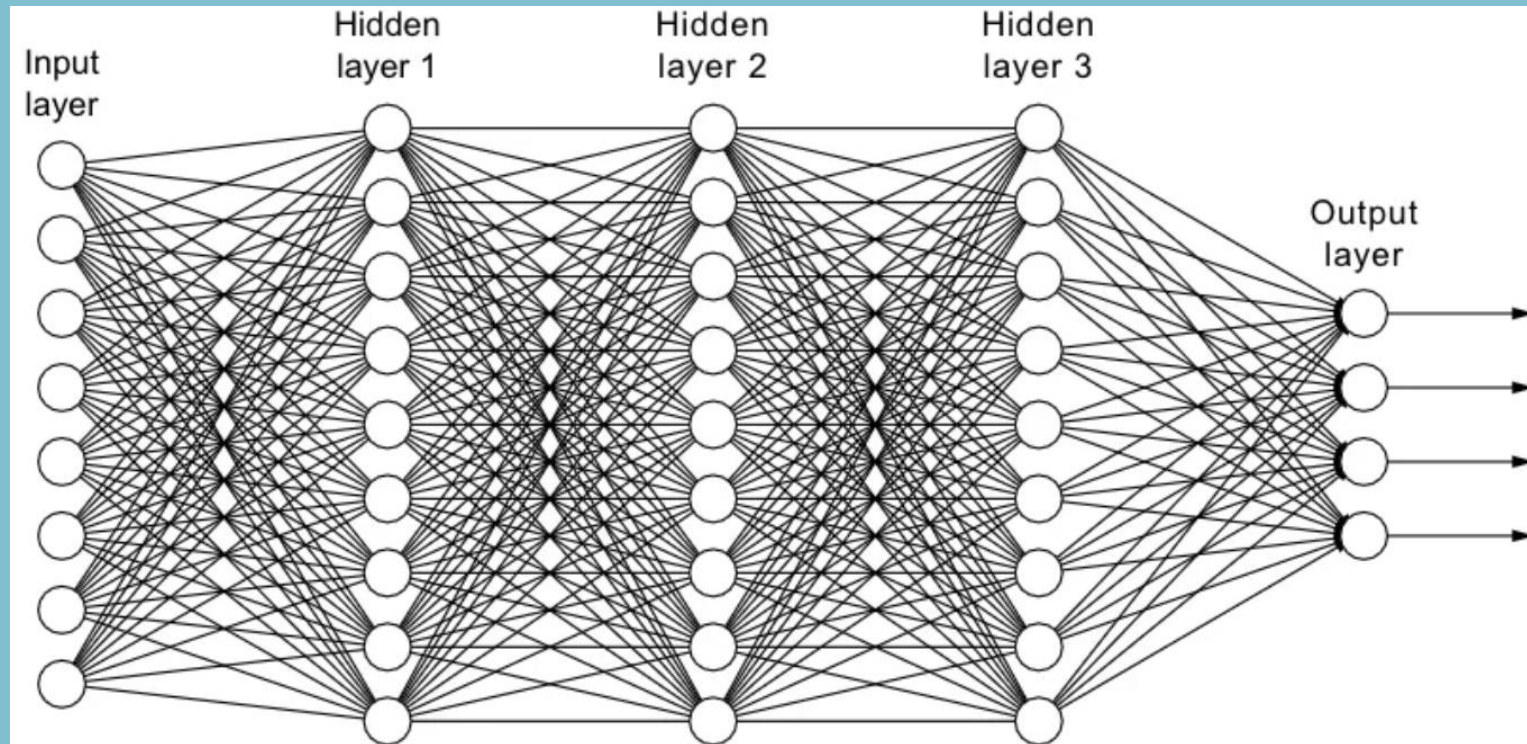
“Nebraska”

?

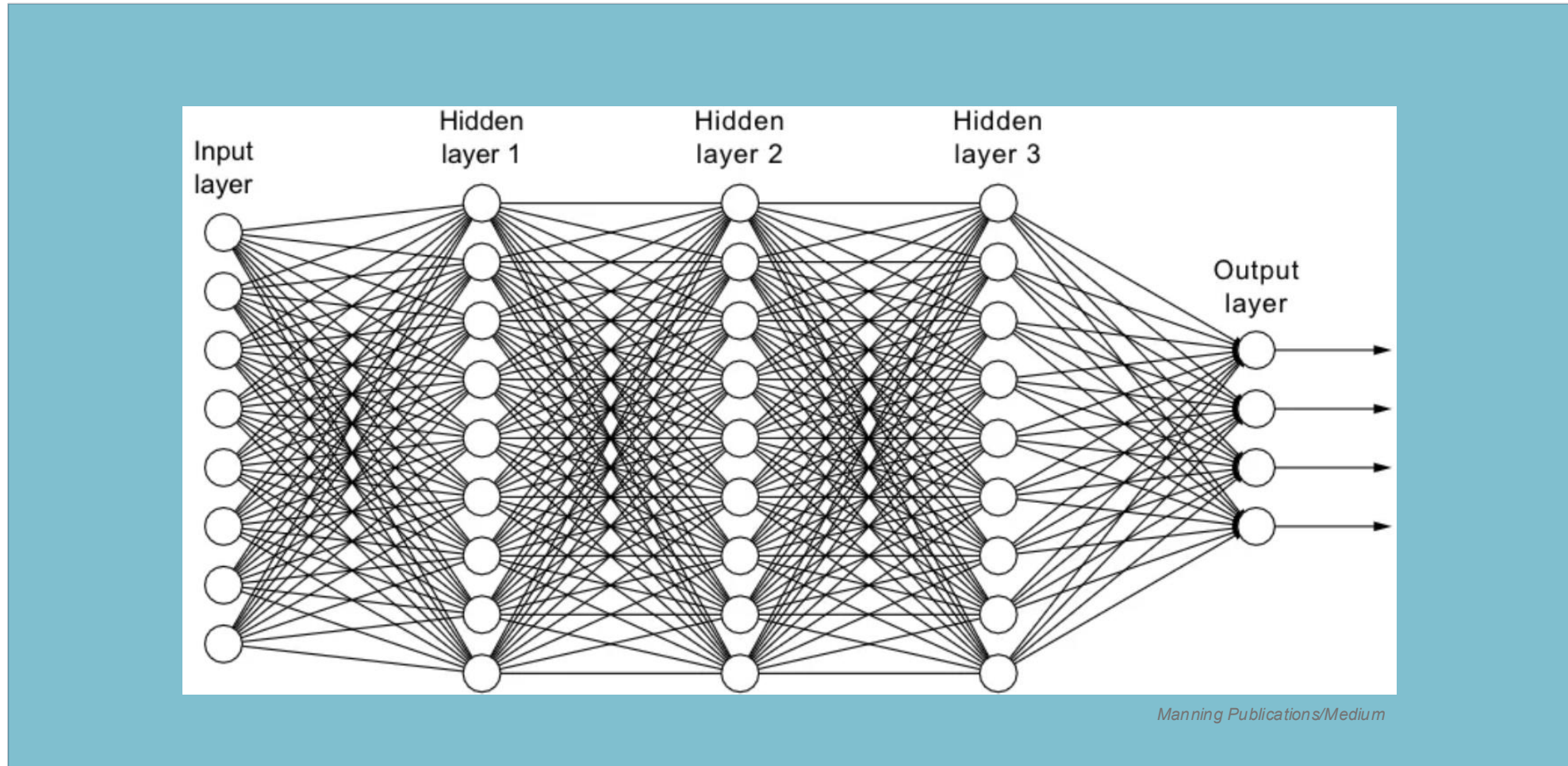
Model

Model

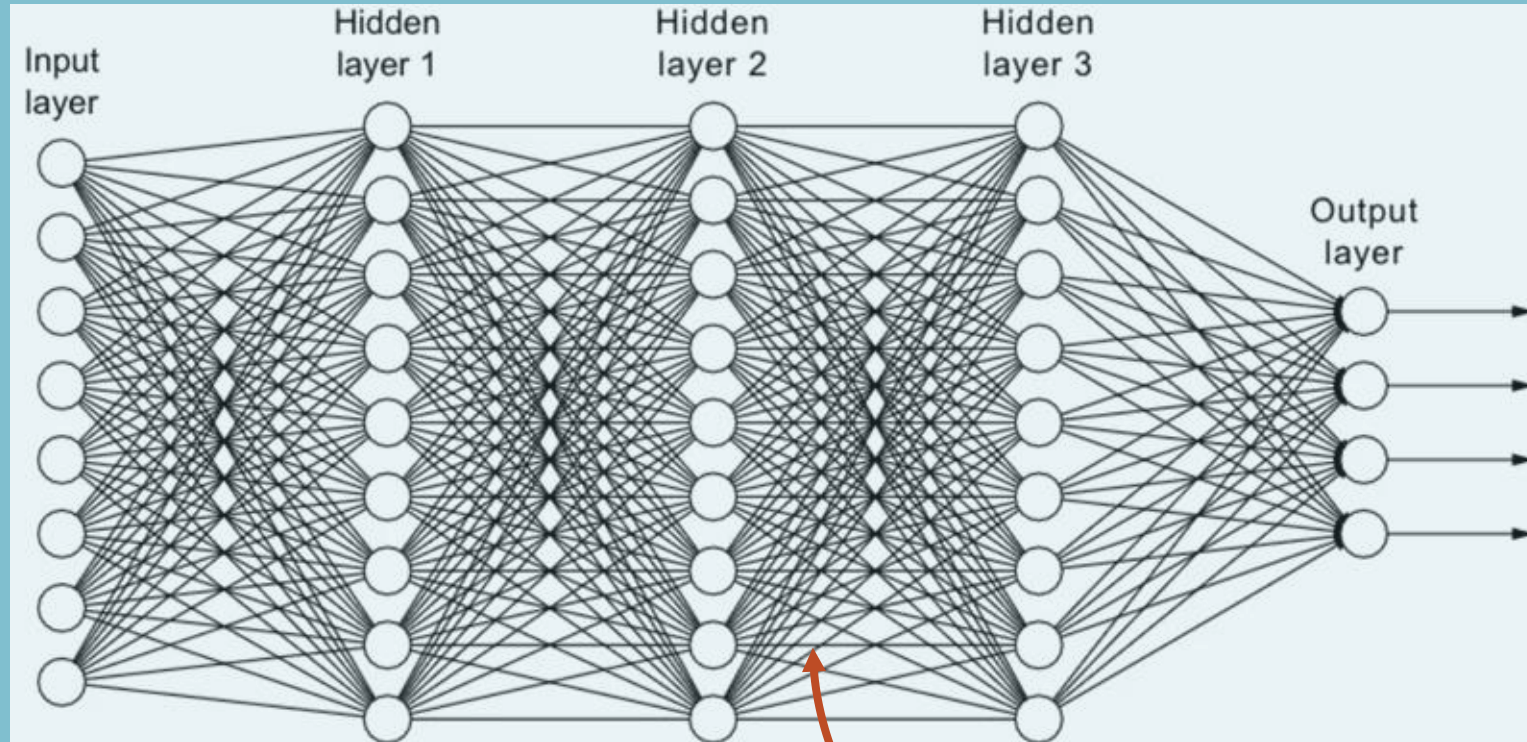
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Manning Publications/Medium

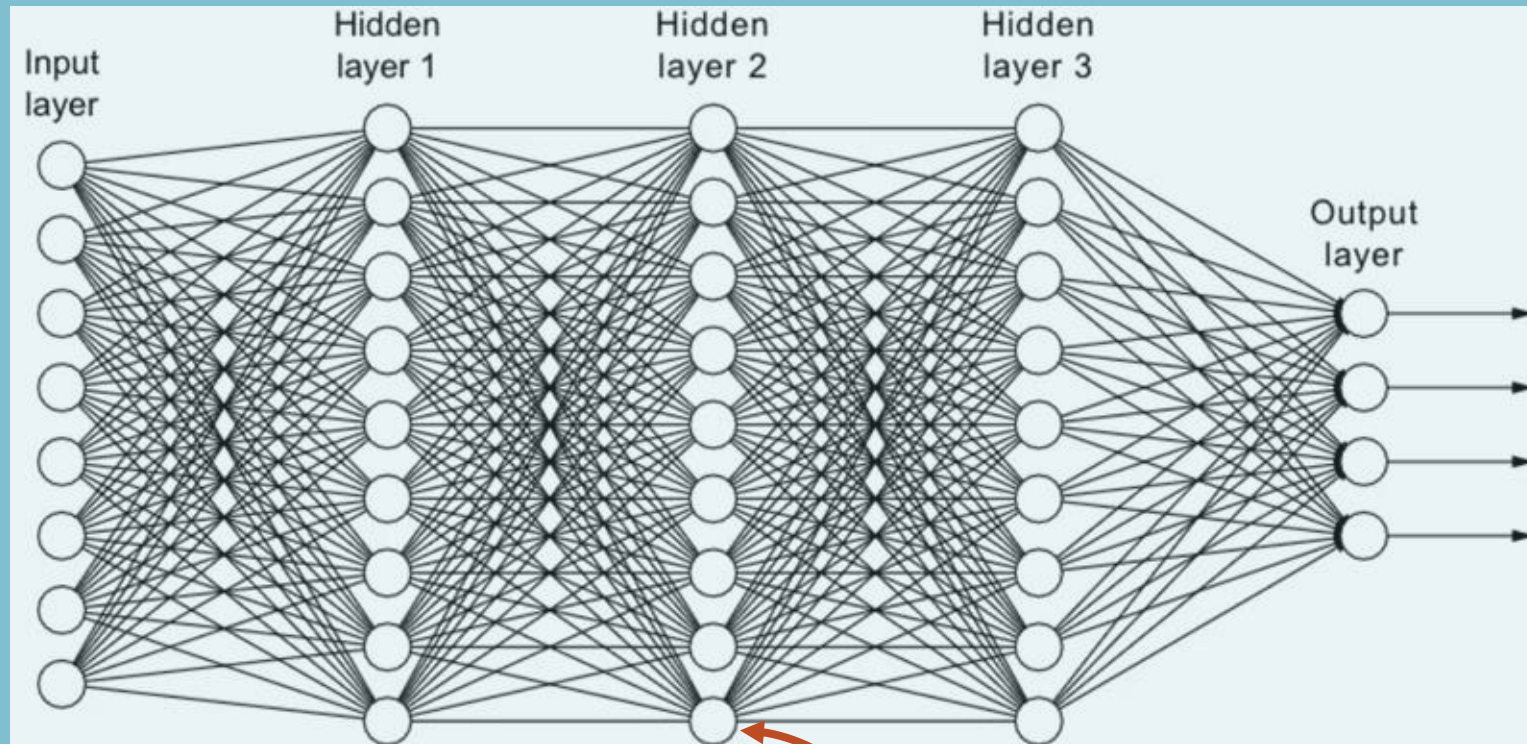


Neural Network



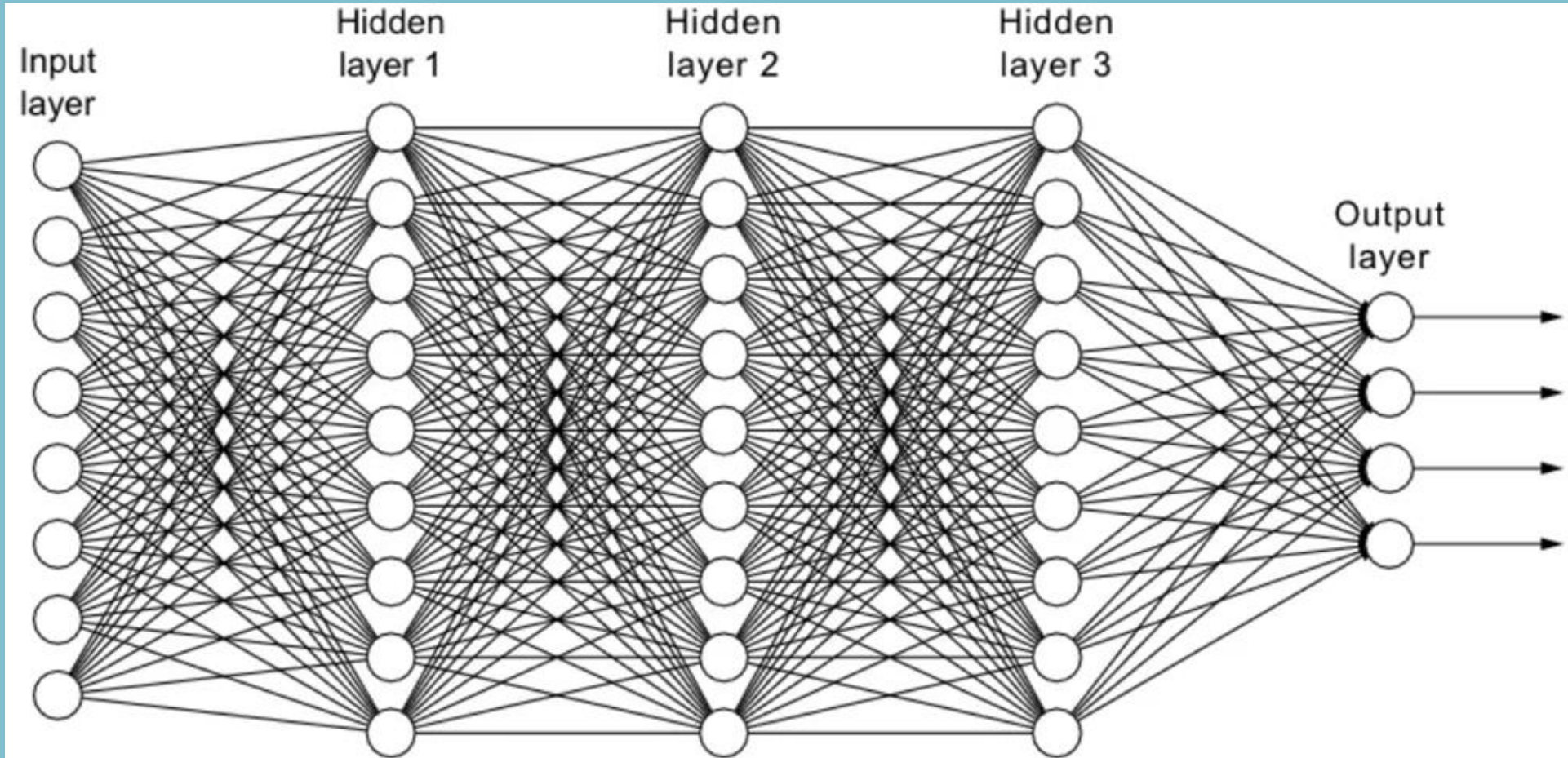
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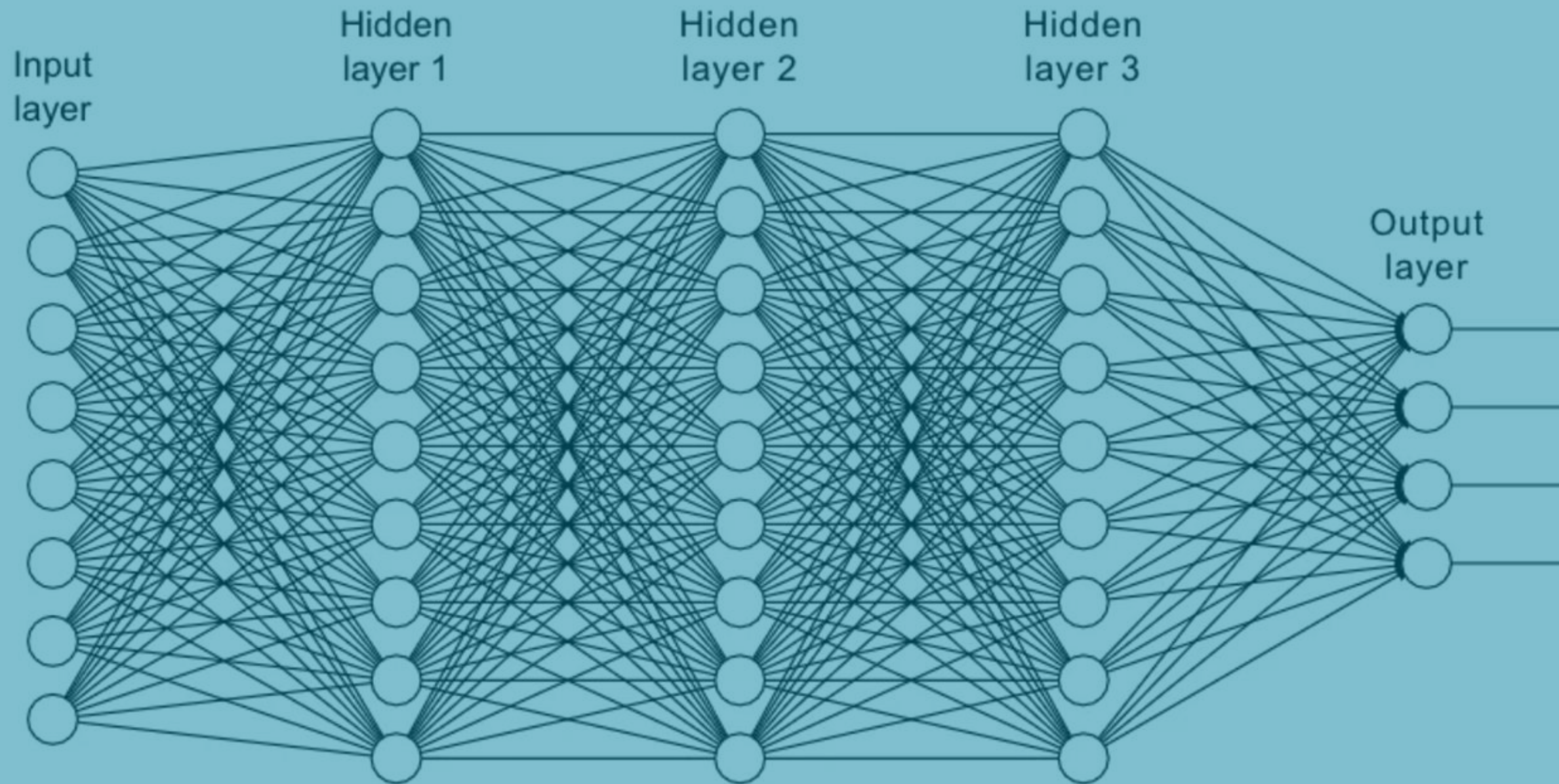
parameters

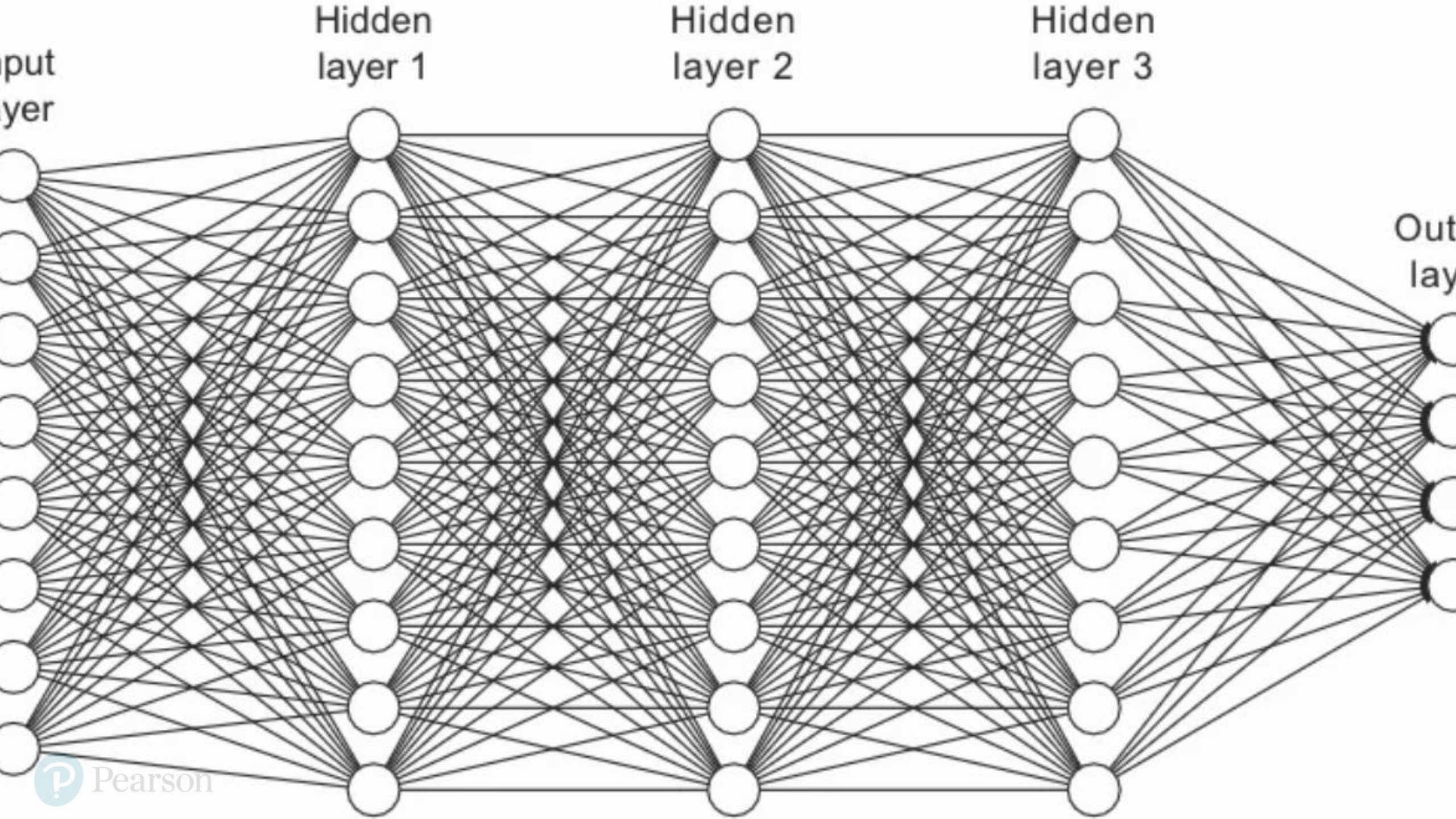


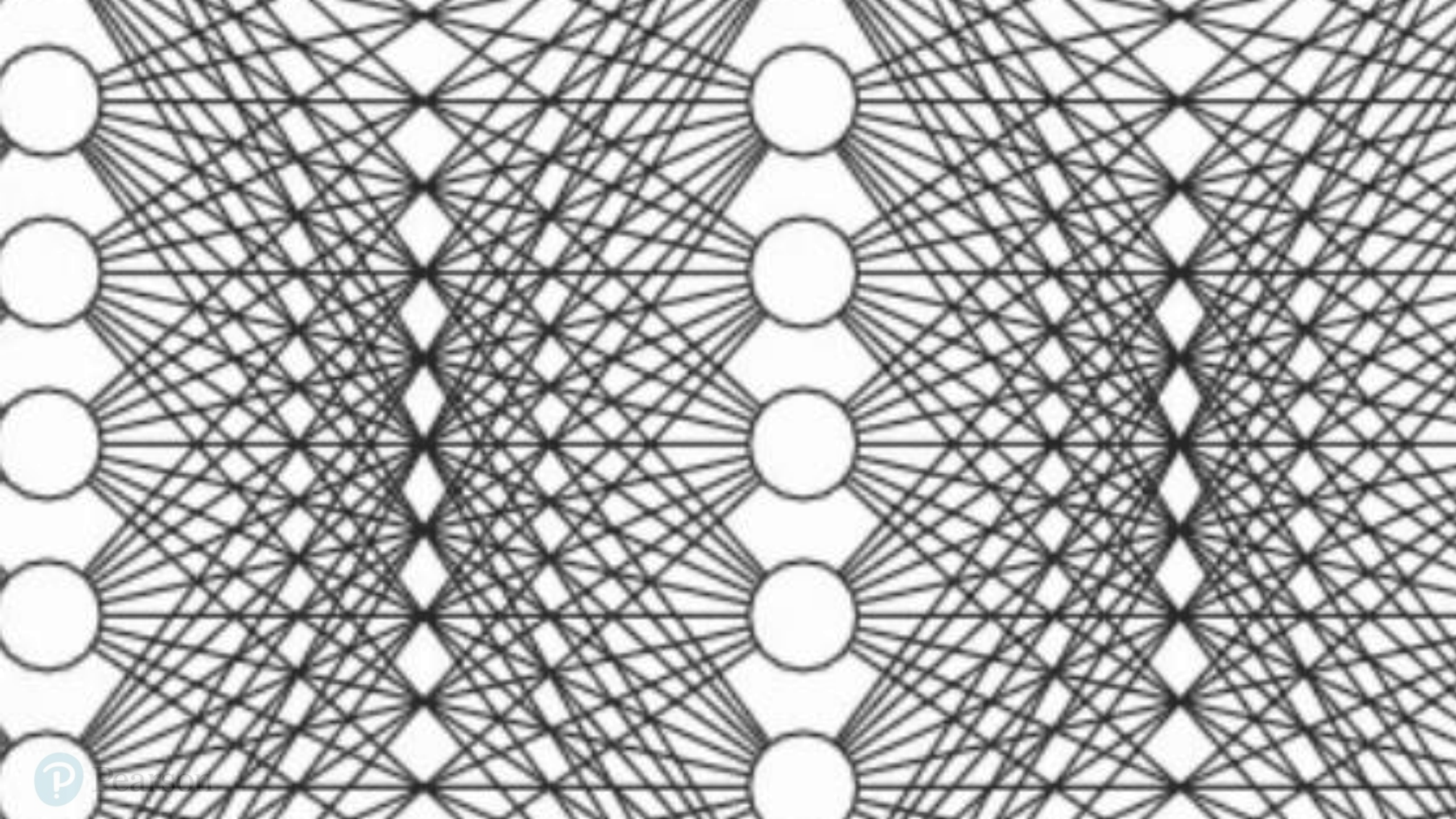
Manning Publications/Medium

neurons

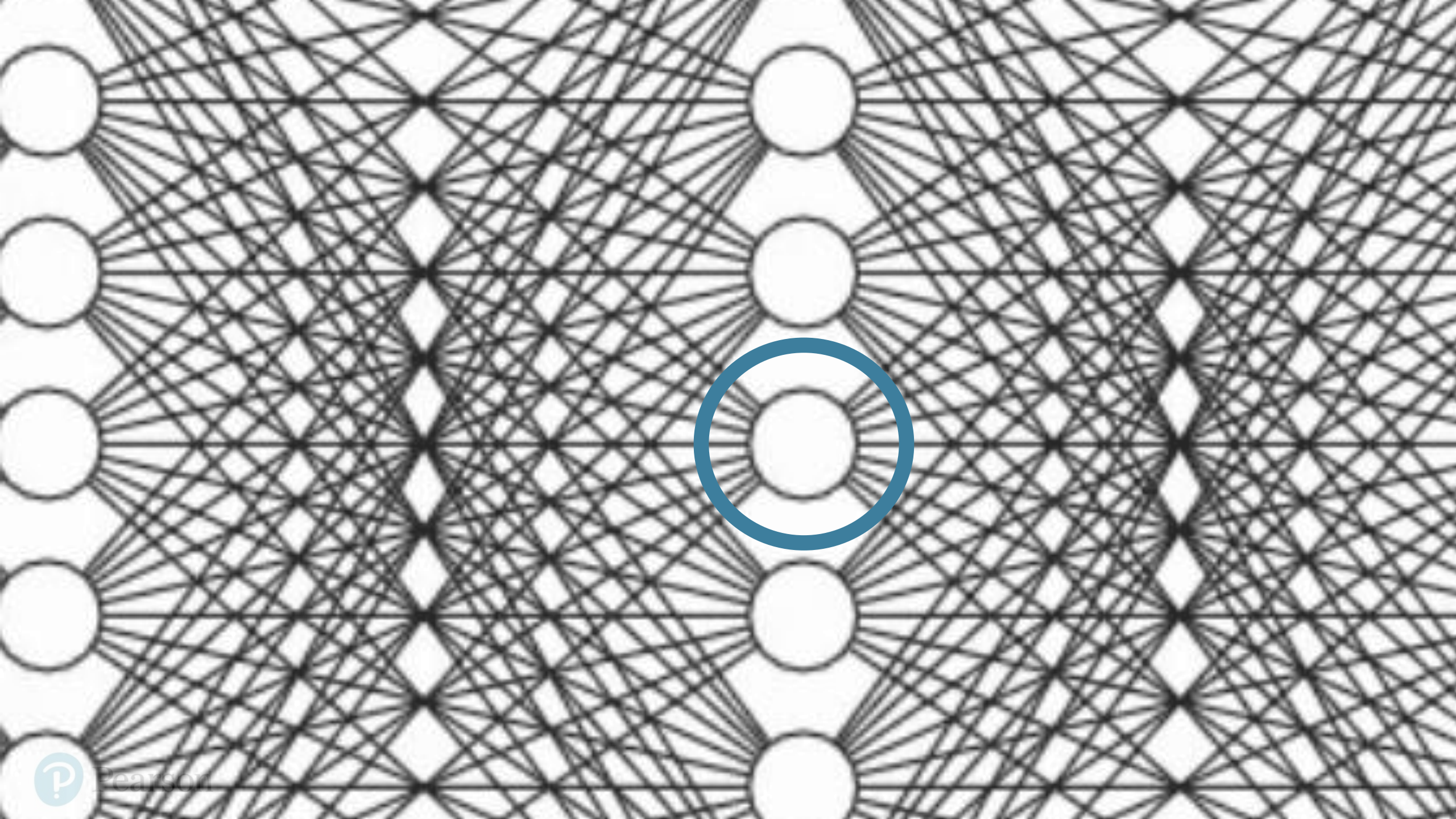


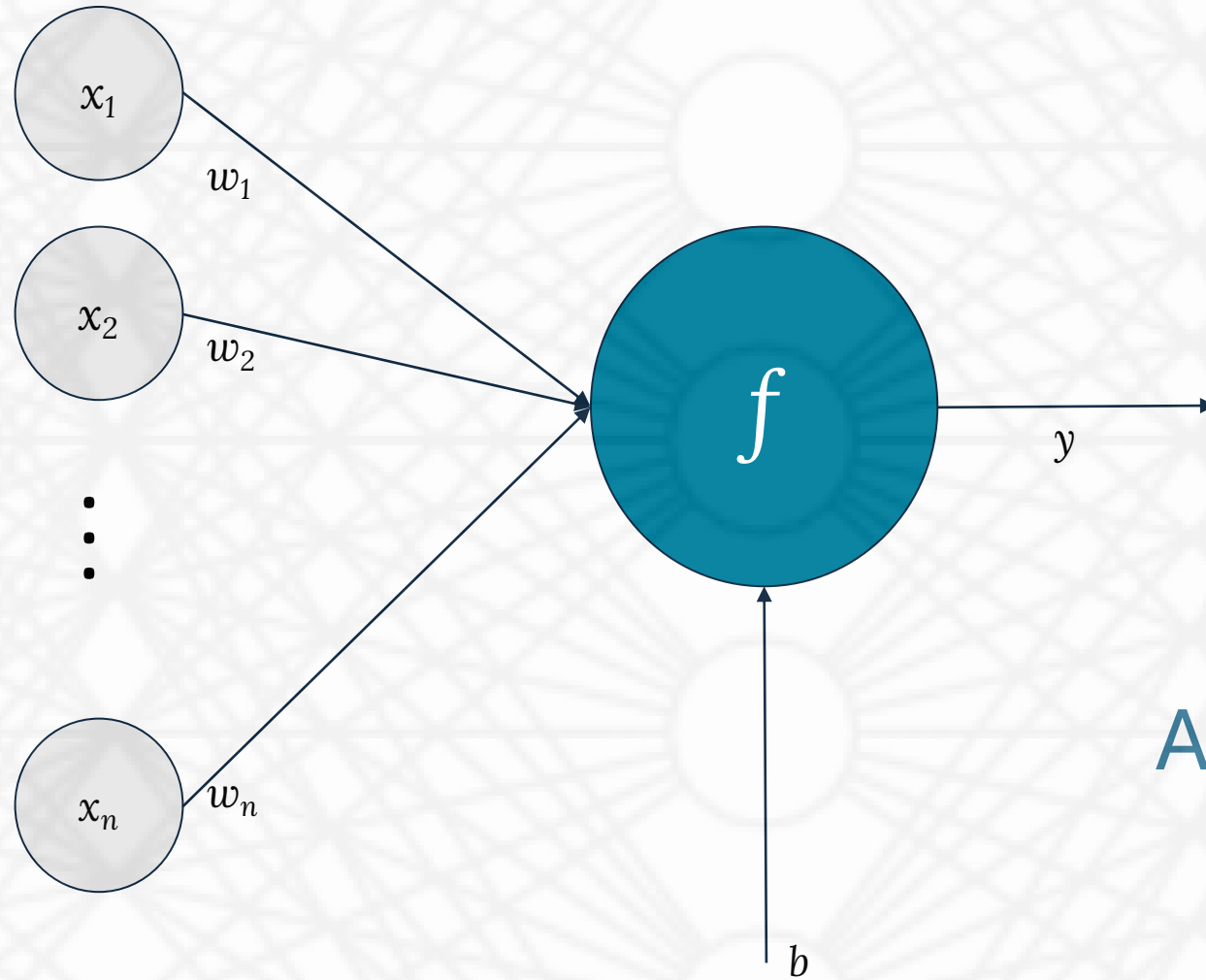






Pearson





A single neuron

A diagram illustrating a linear equation within a neural network context. A blue-bordered rectangle is tilted and contains the equation $f(b + \sum_{i=1}^n x_i w_i)$. The rectangle is connected to several labels: x_1 and w_1 are connected to the first term of the sum; x_2 is connected to the second term; n is connected to the upper limit of the summation; and b is connected to the bias term. The entire diagram is set against a background of faint, overlapping circles.

$$f(b + \sum_{i=1}^n x_i w_i)$$

but actually...

Input

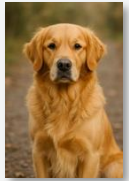


Model



Output

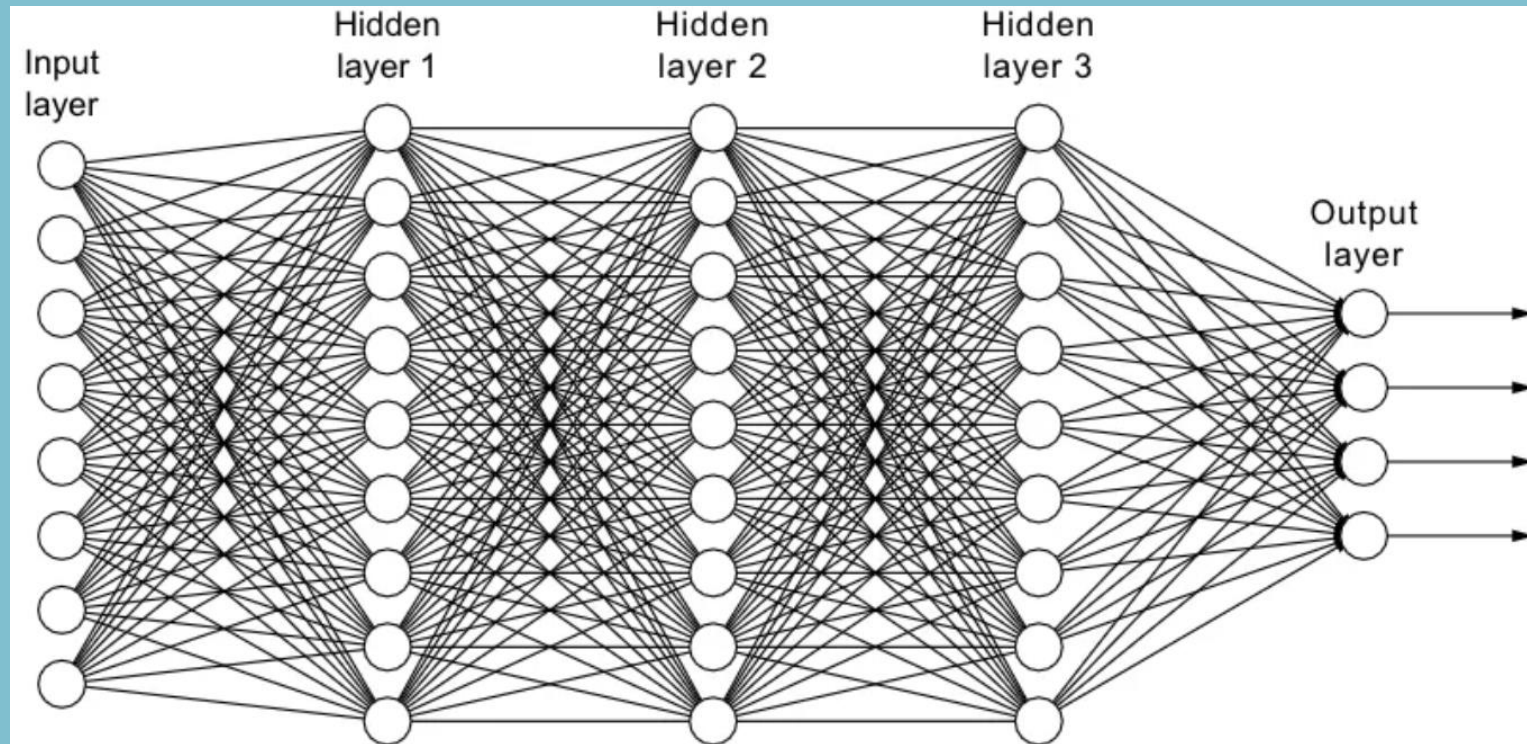
“dog”



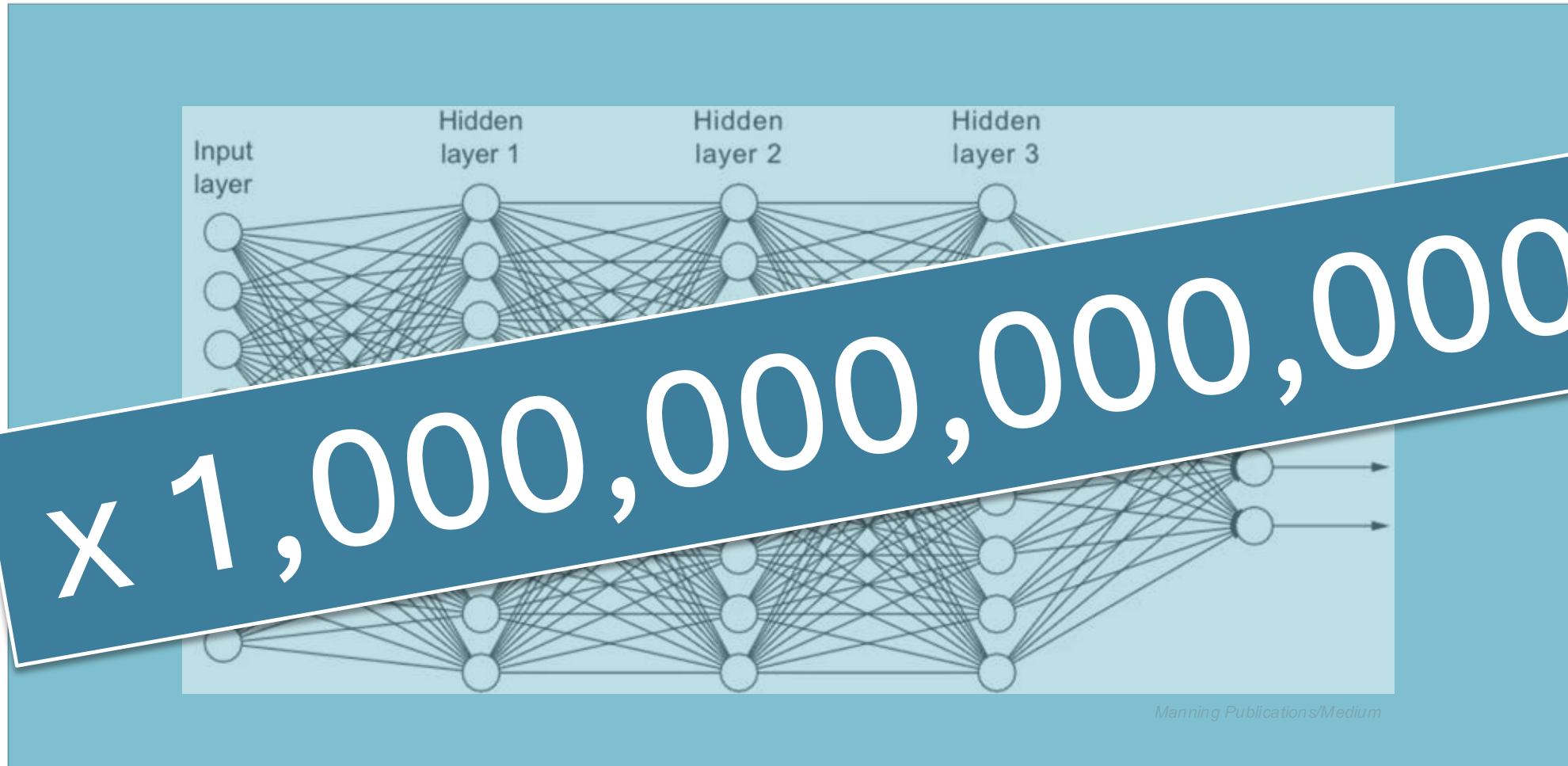
$\text{softmax}(b_6 + w_6 \cdot \text{ReLU}(b_5 + w_5 \cdot \text{ReLU}(b_4 + w_4 \cdot \text{ReLU}(b_3 + w_3 \cdot \text{ReLU}(b_2 + w_2 \cdot \text{ReLU}(b_1 + w_1 \cdot x))))))$



“dog”



Manning Publications/Medium



Neural Network

To fully understand how a **neural network** works, you need:

Matrix Math

$$\begin{bmatrix} 1 & 3 \\ 4 & 2 \\ 0 & 5 \end{bmatrix} + \begin{bmatrix} -2 & 1 \\ 3 & -3 \\ 2 & 4 \end{bmatrix} \times \begin{bmatrix} -11 & 9 \\ 39 & -5 \\ 29 & 13 \end{bmatrix}$$

Mostly matrix multiplication

To fully understand how a **neural network** works, you need:

Matrix Math

$$\begin{bmatrix} 1 & 3 \\ 4 & 2 \\ 0 & 5 \end{bmatrix} + \begin{bmatrix} -2 & 1 \\ 3 & -3 \\ 2 & 4 \end{bmatrix} \times \begin{bmatrix} -11 & 9 \\ 39 & -5 \\ 29 & 13 \end{bmatrix}$$

Mostly matrix multiplication

Simple Calculus

$$\frac{d}{dx} = \sigma'(x) f(ax)$$

Mostly first-order derivatives

To fully understand how a **neural network** works, you need:

Matrix Math

$$\begin{bmatrix} 1 & 3 \\ 4 & 2 \\ 0 & 5 \end{bmatrix} + \begin{bmatrix} -2 & 1 \\ 3 & -3 \\ 2 & 4 \end{bmatrix} \times \begin{bmatrix} -11 & 9 \\ 39 & -5 \\ 29 & 13 \end{bmatrix}$$

Mostly matrix multiplication

Simple Calculus

$$\frac{d}{dx} = \sigma'(x) f(ax)$$

Mostly first-order derivatives

Simple Probability

$$P(y_i \mid \mathbf{x}) = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}$$

Conditional probability, etc.

Audience Poll Question

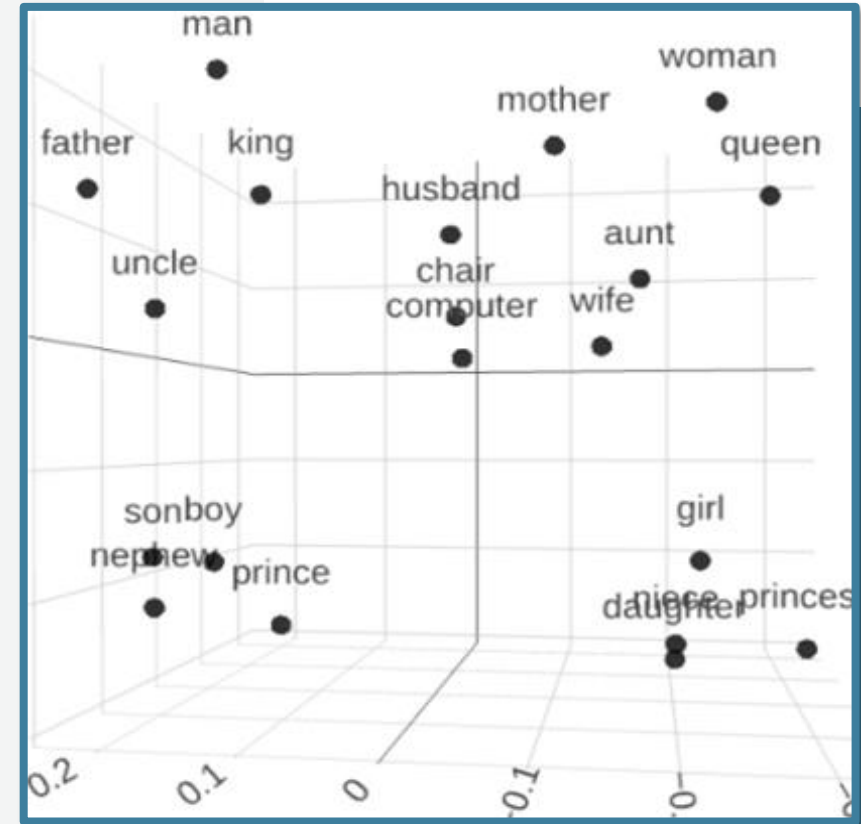
What is the best definition of **machine learning** out of the following choices? (single choice)

- The attempt to create a robot that can do everything a human can do
- Static, deterministic classification algorithms applied to input data
- Methods that allow algorithms to learn to recognize patterns in input data
- Humans “teaching” computers to do a task by giving instructions

Section 2: Word Embeddings

By the end of this section you will know:

- What word embeddings are
- How they are formed
- Why we need them
- What we can do with them
- What the image on the right means



Humans and Language

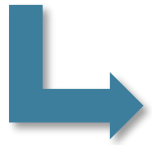


Our understanding and use of language is unique and unquestionably complicated

Humans and Language



Our understanding and use of language is unique and unquestionably complicated



So how do we somehow capture this complicated concept in a model?

(Brief) History of Natural Language Processing

Pre-Machine Learning

1960s

Hard coded rules for very specific tasks

```
Welcome to

EEEEEE LL      IIII ZZZZZZ  AAAAA
EE      LL      II      ZZ  AA  AA
EEEEEE LL      II      ZZ  AAAAAA
EE      LL      II      ZZ  AA  AA
EEEEEE LLLLLL IIII ZZZZZZ  AA  AA

Eliza is a mock Rogerian psychotherapist.
The original program was described by Joseph Weizenbaum in 1966.
This implementation by Norbert Landsteiner 2005.

ELIZA: Is something troubling you ?
YOU:   Men are all alike.
ELIZA: What is the connection, do you suppose ?
YOU:   They're always bugging us about something or other.
ELIZA: Can you think of a specific example ?
YOU:   Well, my boyfriend made me come here.
ELIZA: Is it important to you that your boyfriend made you come here ?
YOU:   He says I'm depressed much of the time.
ELIZA: I am sorry to hear that you are depressed.
YOU:   It's true. I am unhappy.
ELIZA: Can you explain what made you unhappy ?
YOU:
```

(Brief) History of Natural Language Processing

Pre-Machine Learning

1960s

```
Welcome to
EEEEEE LL      IIII ZZZZZZ AAAAA
EE      LL      II      ZZ AA  AA
EEEEEE LL      II      ZZZ AAAAAA
EE      LL      II      ZZ AA  AA
EEEEEE LLLLLL IIII ZZZZZZ AA  AA

Eliza is a mock Rogerian psychotherapist.
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YOU:
```

Hard coded rules for very specific tasks

(Brief) History of Natural Language Processing

Pre-Machine Learning

1960s

Hard coded rules for very specific tasks

word2vec

2013

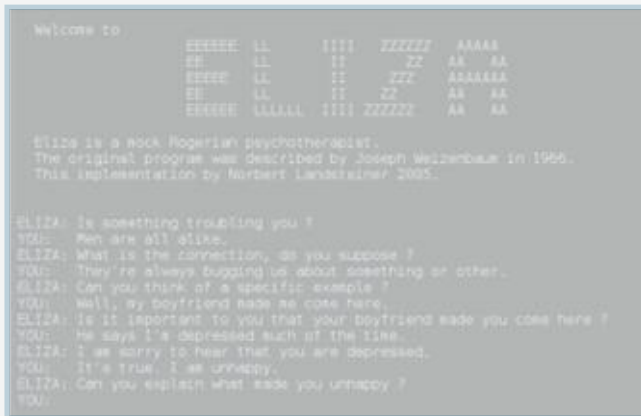


Learned, meaningful, scalable
representation of language

(Brief) History of Natural Language Processing

Pre-Machine Learning

1960s



Hard coded rules for very specific tasks

word2vec

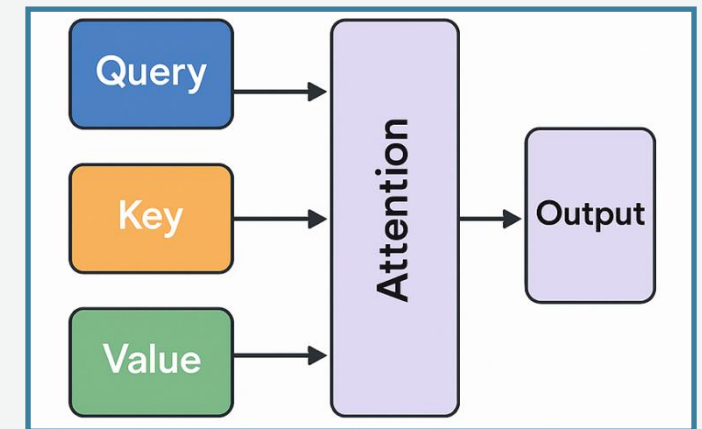
2013



Learned, meaningful, scalable representation of language

Transformer with Attention

2017



More sophisticated language models

How AI Models “Read” Input

It was the best of
times, it was the
worst of times...



Model

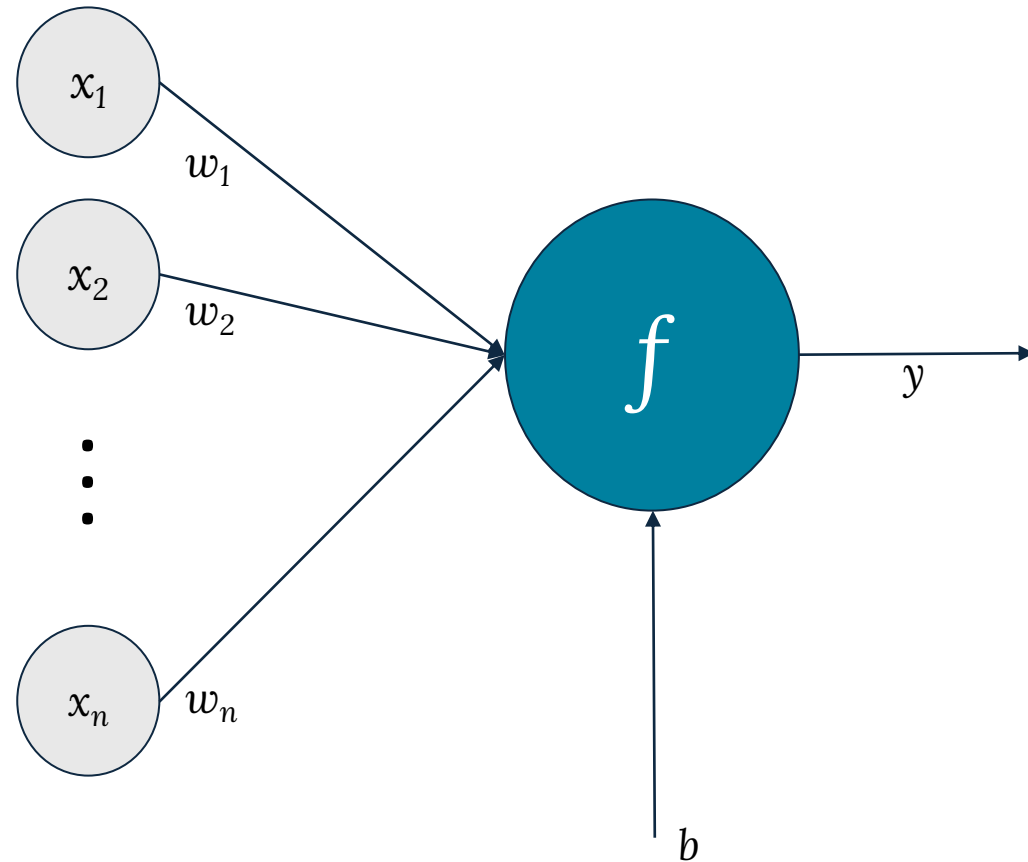
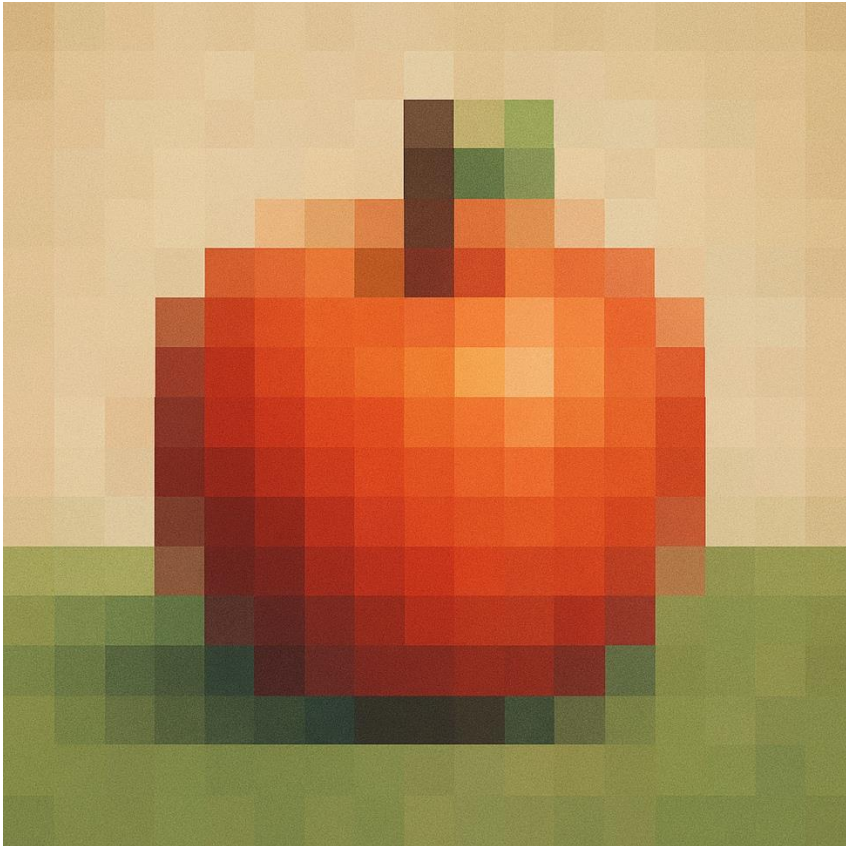
How AI Models “Read” Input

It was the best of
times, it was the
worst of times...

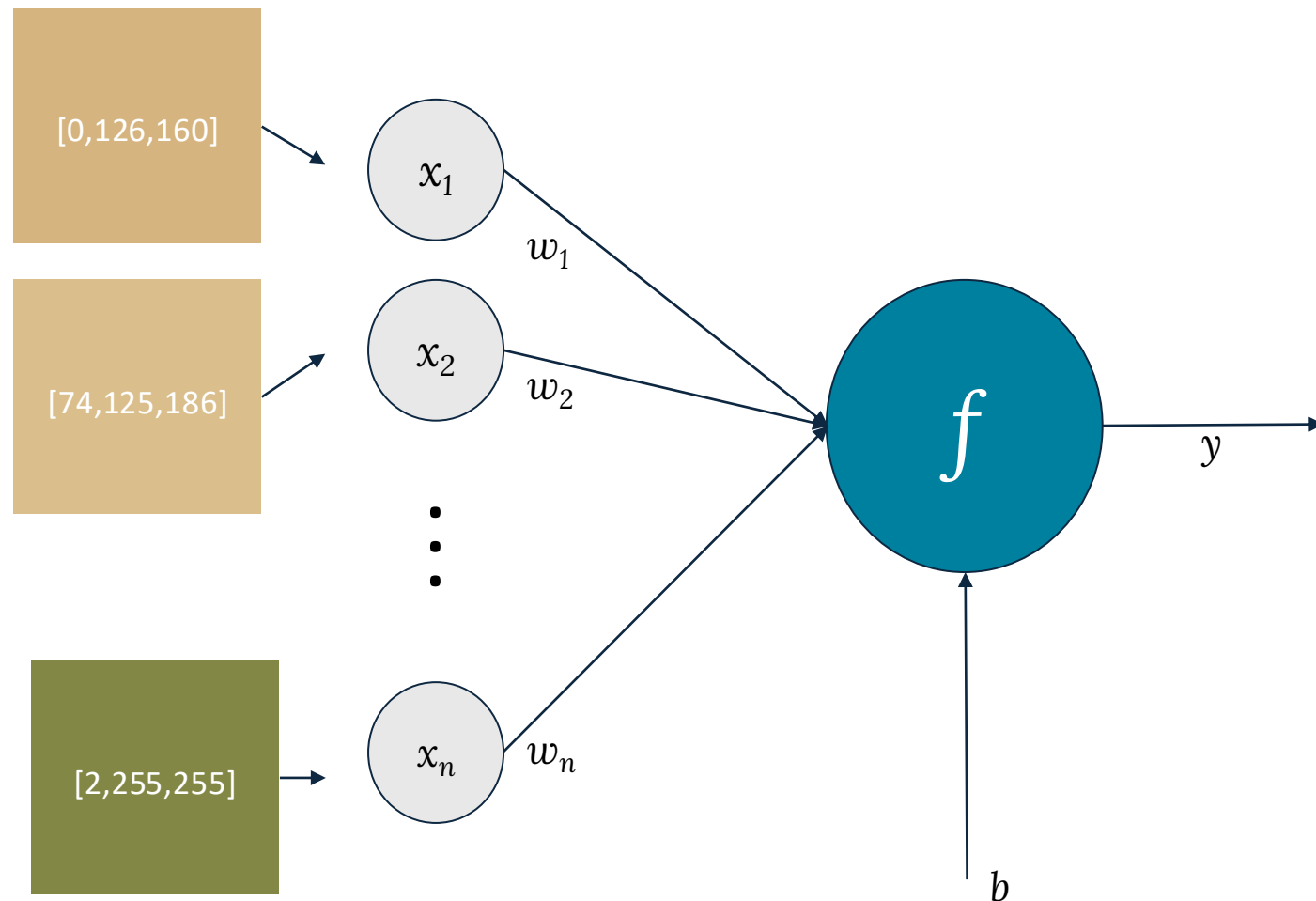


$$f\left(b + \sum_{i=1}^n x_i w_i\right)$$

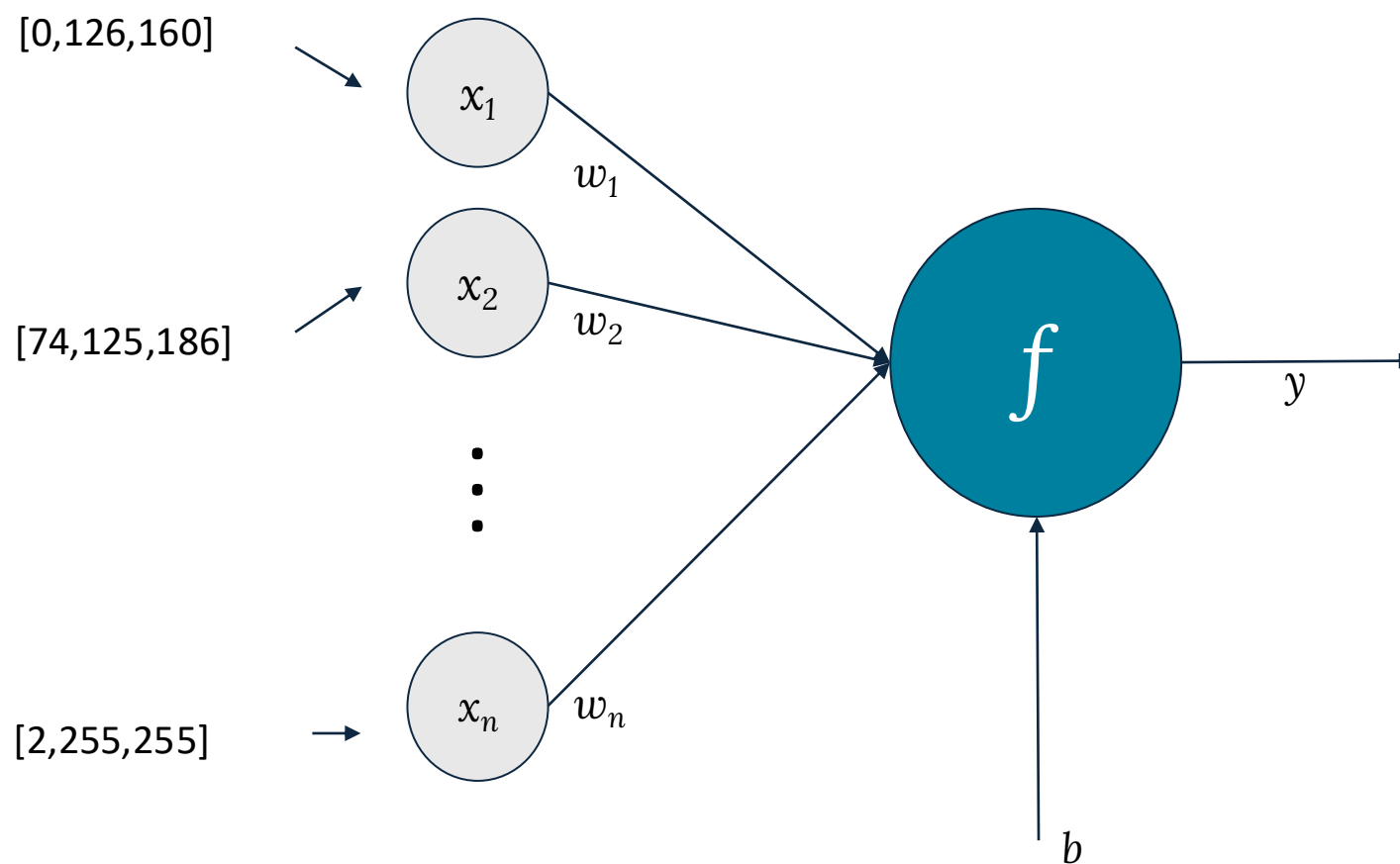
Input for an Image Model



Input for an Image Model



Input for an Image Model



Input for a Language Model?

- What are the “units of meaning” in human language?

Input for a Language Model?

- What are the “units of meaning” in human language?
 - Letters?
 - Syllables?
 - Words?
 - Sentences?
 - Something else?

Audience Poll Question

- What are the “units of meaning” in human language? (single choice)
 - Letters?
 - Syllables?
 - Words?
 - Sentences?
 - Something else?

Input for a Language Model?

For the purposes of modern language modeling, the smallest units of meaning in language are considered to be **words** (or **parts of words**)

Where do words draw their meaning from?

“You shall know a word by the company it keeps!”

— “A synopsis of linguistic theory 1930-1955”, in *Studies in Linguistic Analysis 1930-1955*, 1962.

“After running the marathon, my legs were **wappered**.”

Audience Poll Question

What is the most likely definition of **wappered** from the following choices? (single choice)

- shaking
- tired
- hurting
- orange
- hairy

“After running the marathon, my legs were **wappered**.”

“I need to go to sleep; I’m **wappered** out.”

Audience Poll Question

What is the most likely definition of **wappered** from of the following choices? (single choice)

- shaking
- tired
- hurting
- orange
- hairy

"She owned a pet _____."

A) dog

B) cat

C) elephant

D) haircut

"She owned a pet ____."

- A) dog
- B) cat
- C) elephant
- D) haircut

Puddles - In Lacey's Petshop, it is revealed that when Lacey was younger, she owned a pet dog named "Puddles" that she was very fond of. Unfortunately, Puddles ended up being murdered by a couple weeks, possibly for biting him.

has history with Scooby. She met him at a dog show in the episode. In real life, Wanda Sykes has a thing for animals. She owned a pet dog called Riley. She was at an actual dog show. Her comedy routines involve pet jokes. She has also provided the voice of animal

But the call handler just asked scripted questions – including her husband's ethnicity and if she owned a pet dog.

"She owned a pet dog"

"She owned a pet ____."

- A) dog
- B) cat
- C) elephant
- D) haircut

She owned a pet cat called Laith. She loved painting the cat and enough cat portraits.^[8] She also enjoyed experimenting to create

struggled to ride her horse for the prior three months. She had lived in France her whole life, and she'd never traveled abroad before. She owned a pet cat and she had recently been in contact with cattle, the journal noted.

- She owned a pet cat named Rui, who died on 23 September 2021.
- She used a slightly different voice at debut, but stopped it after it

"She owned a pet cat"

"She owned a pet _____."

- A) dog
- B) cat
- C) elephant
- D) haircut

It's awesome Betty loved animals so much. I just animals. **She** in fact **owned** an **elephant**. 2 yrs.

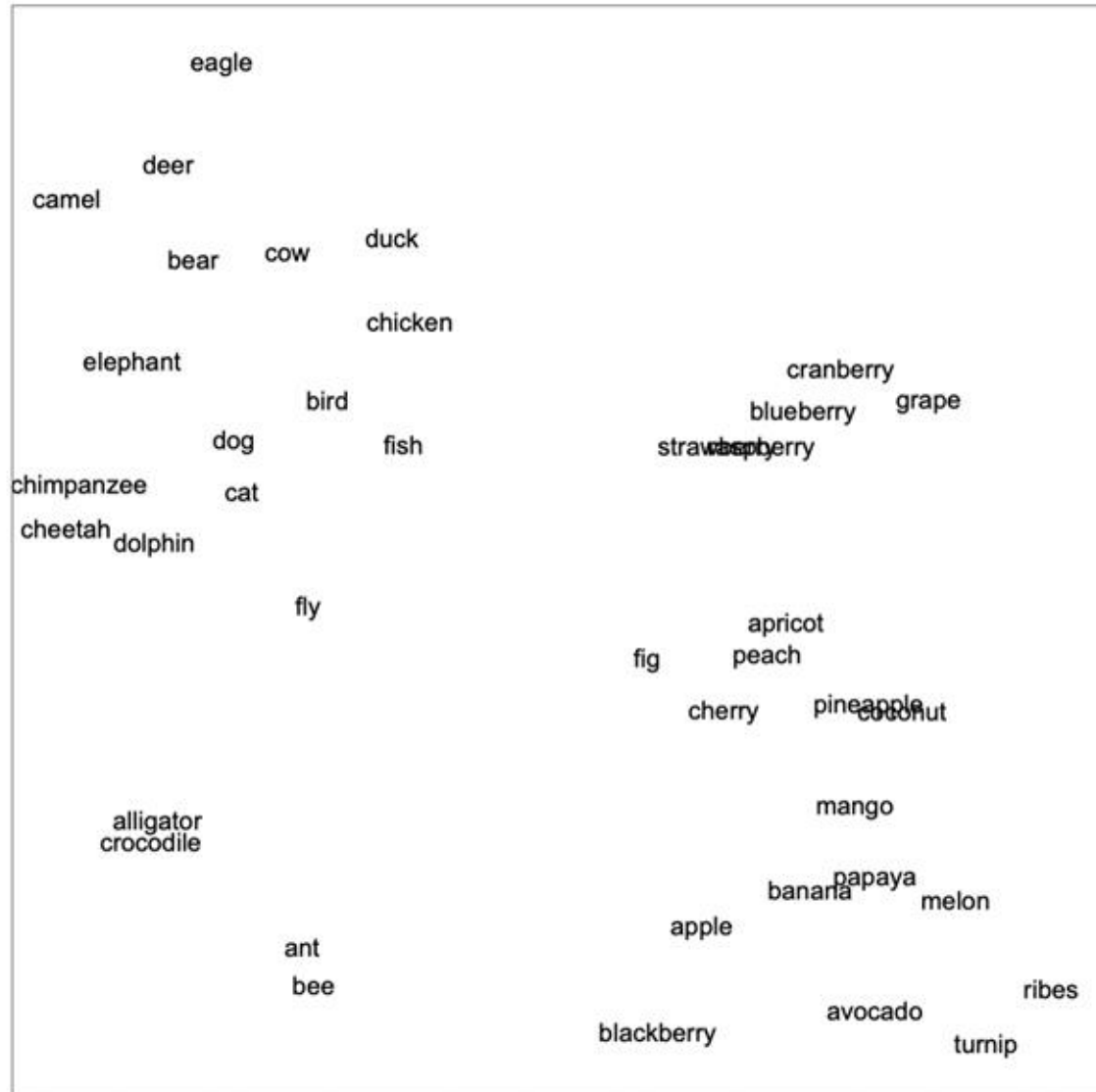
"She owned a pet elephant"

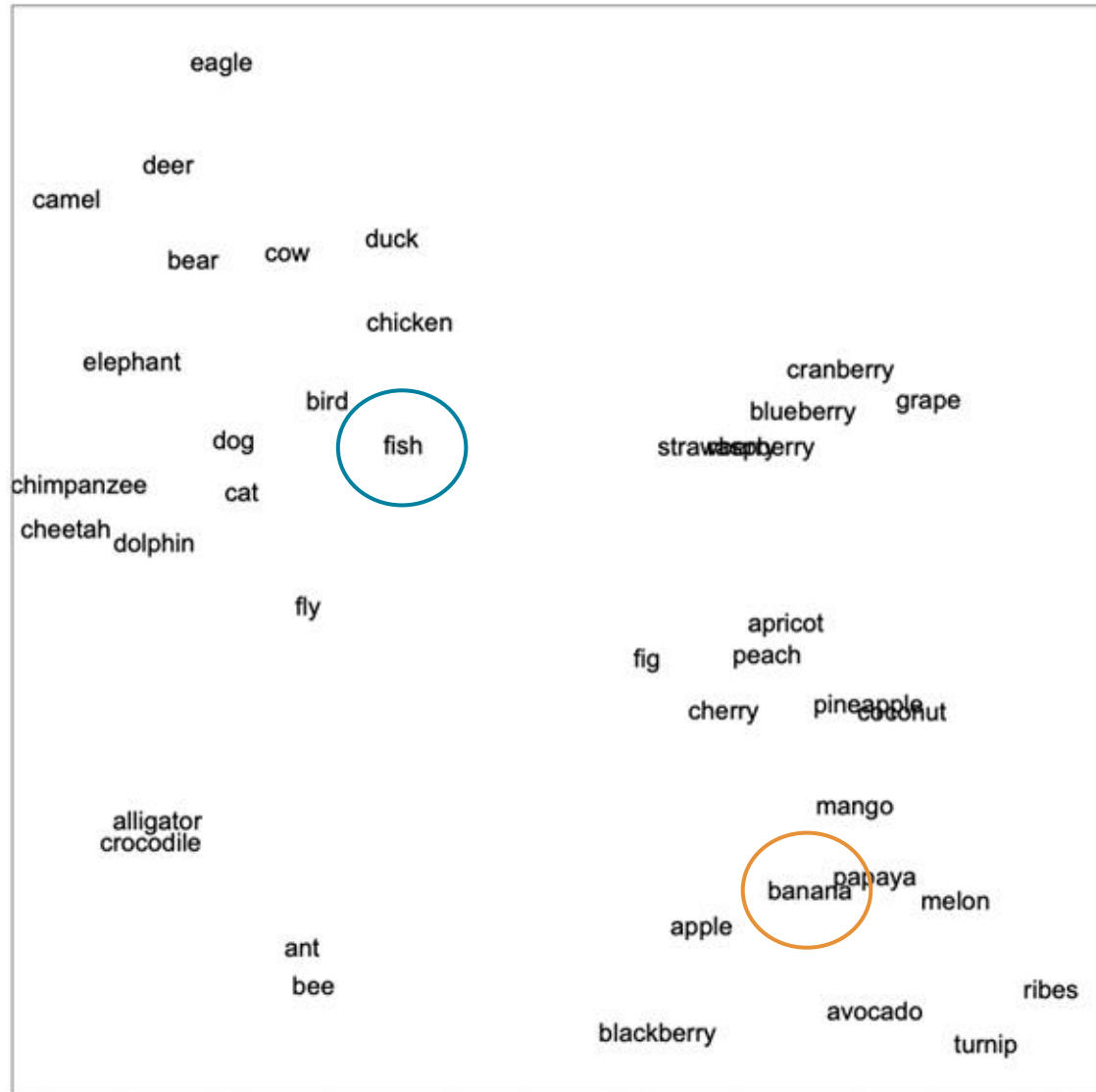
"She owned a pet ____."

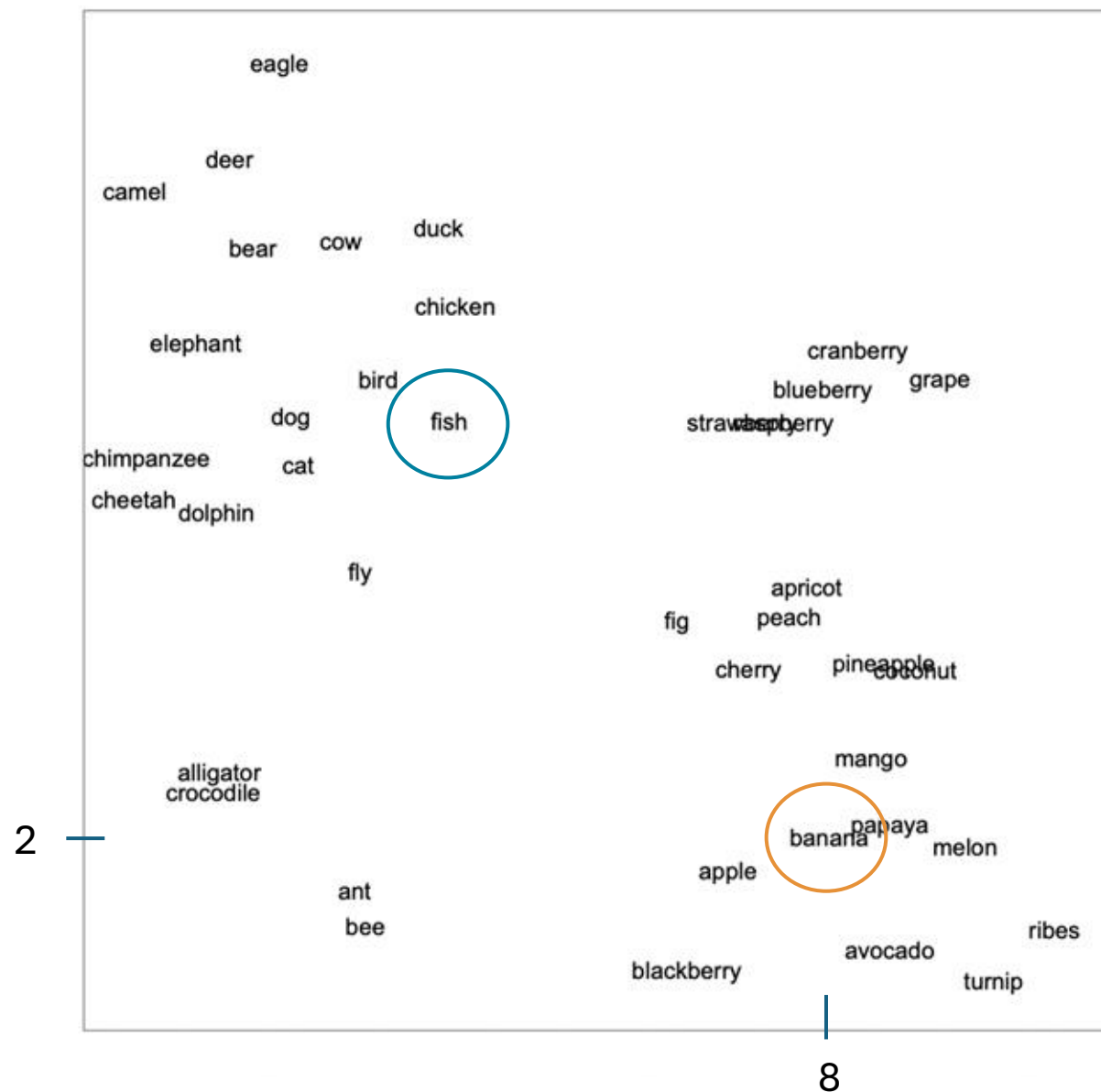
- A) dog
- B) cat
- C) elephant
- D) haircut

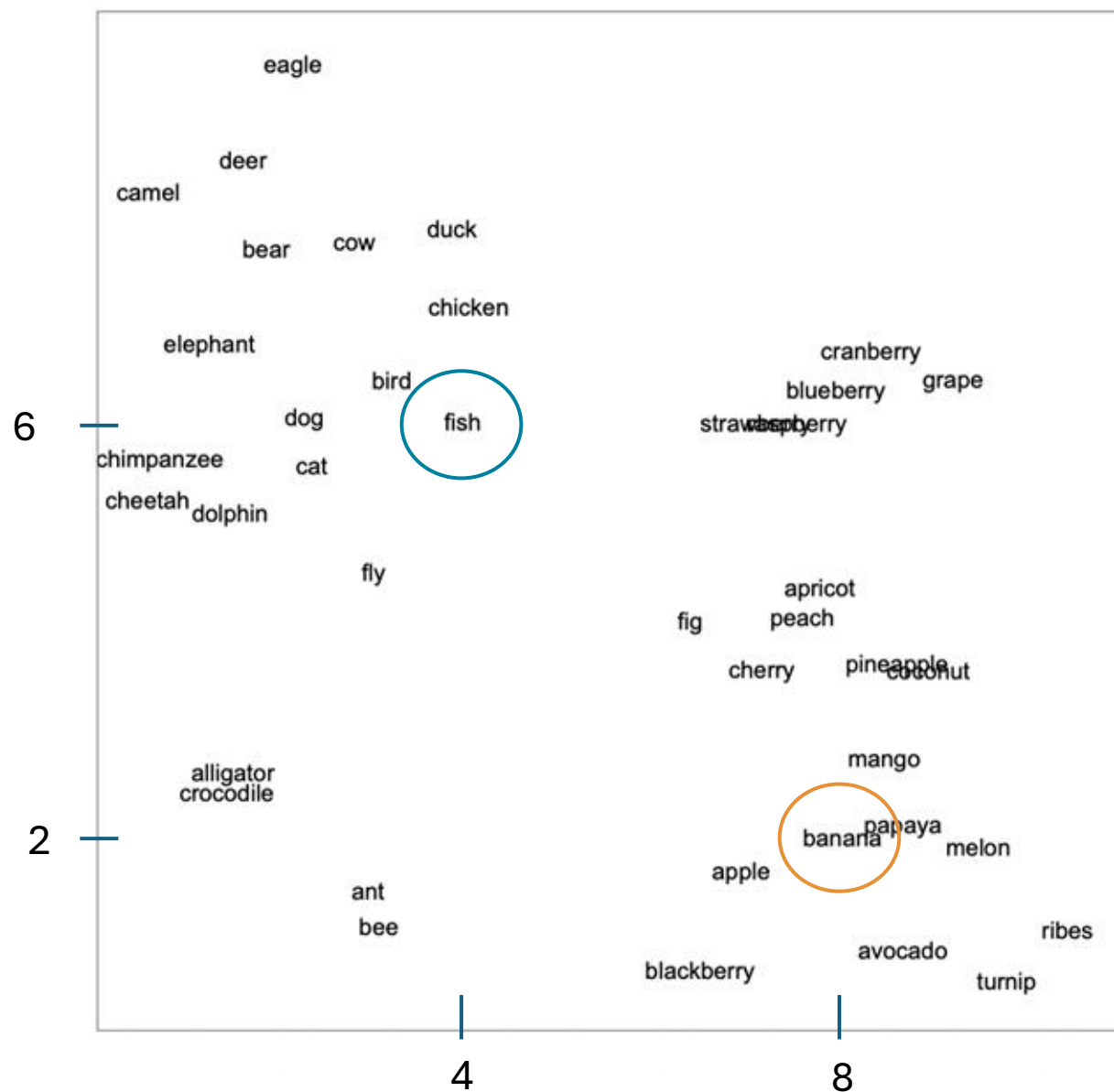


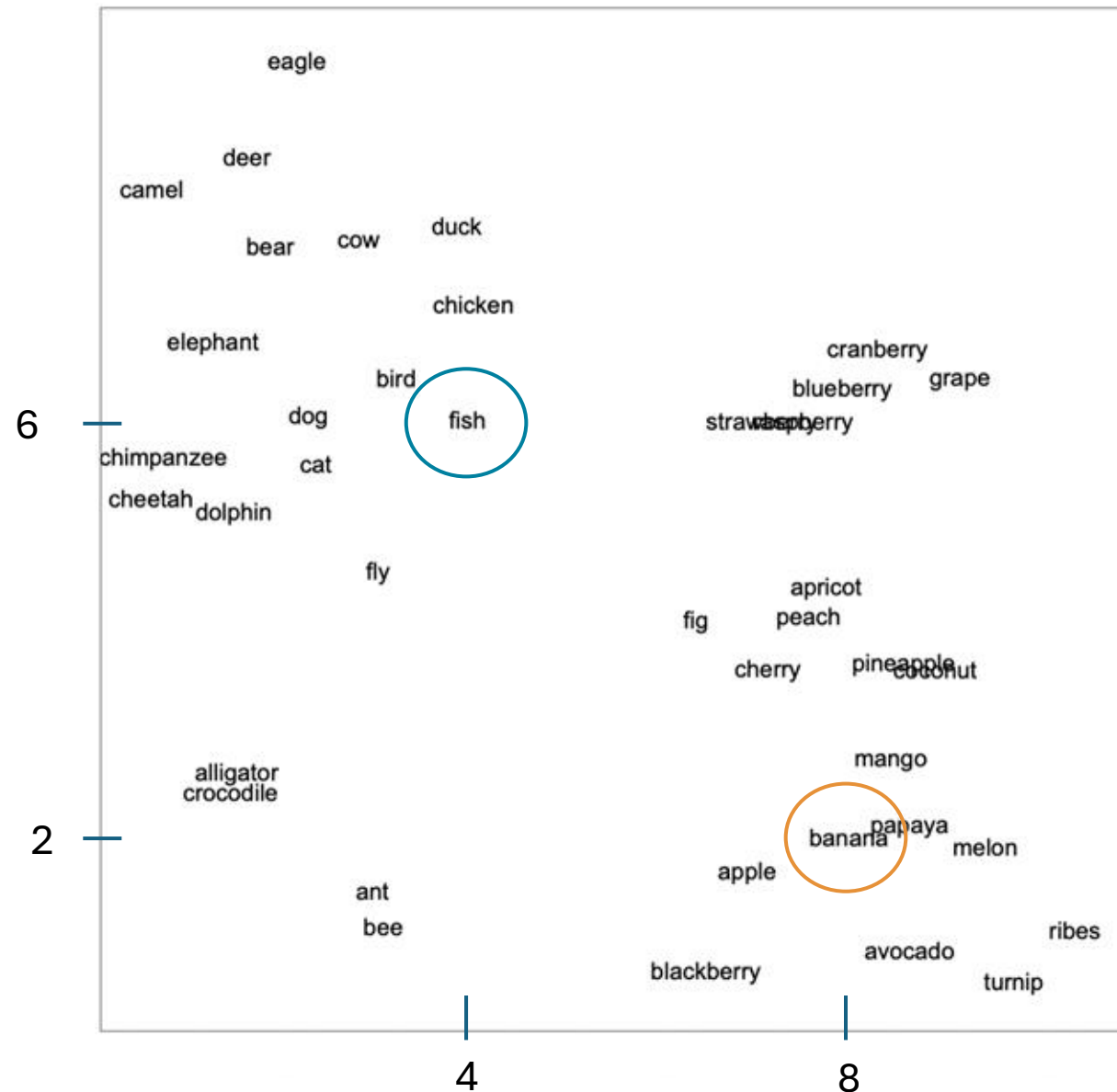
"She owned a pet haircut"











fish = [4,6]

banana = [8,2]

?

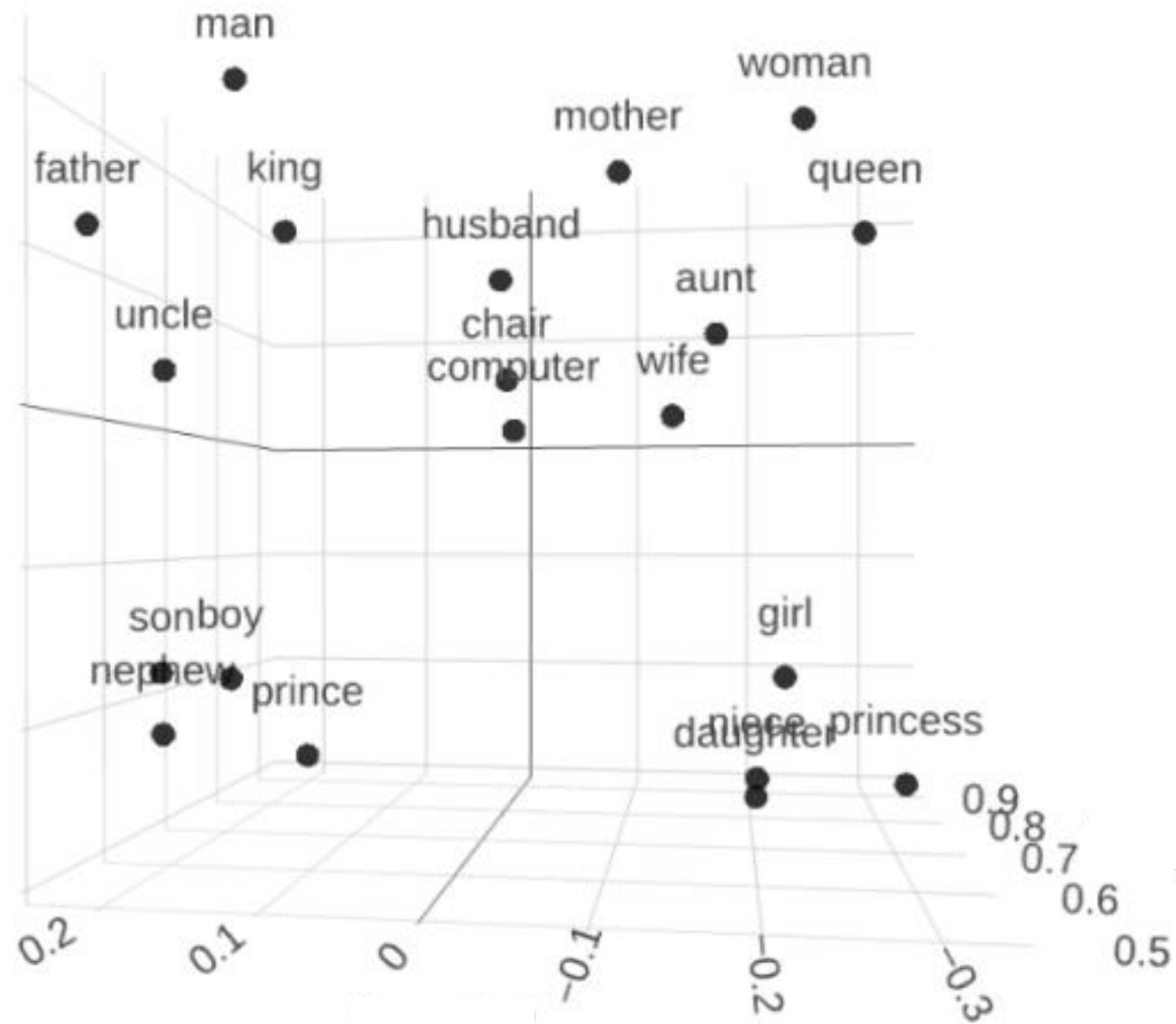
1

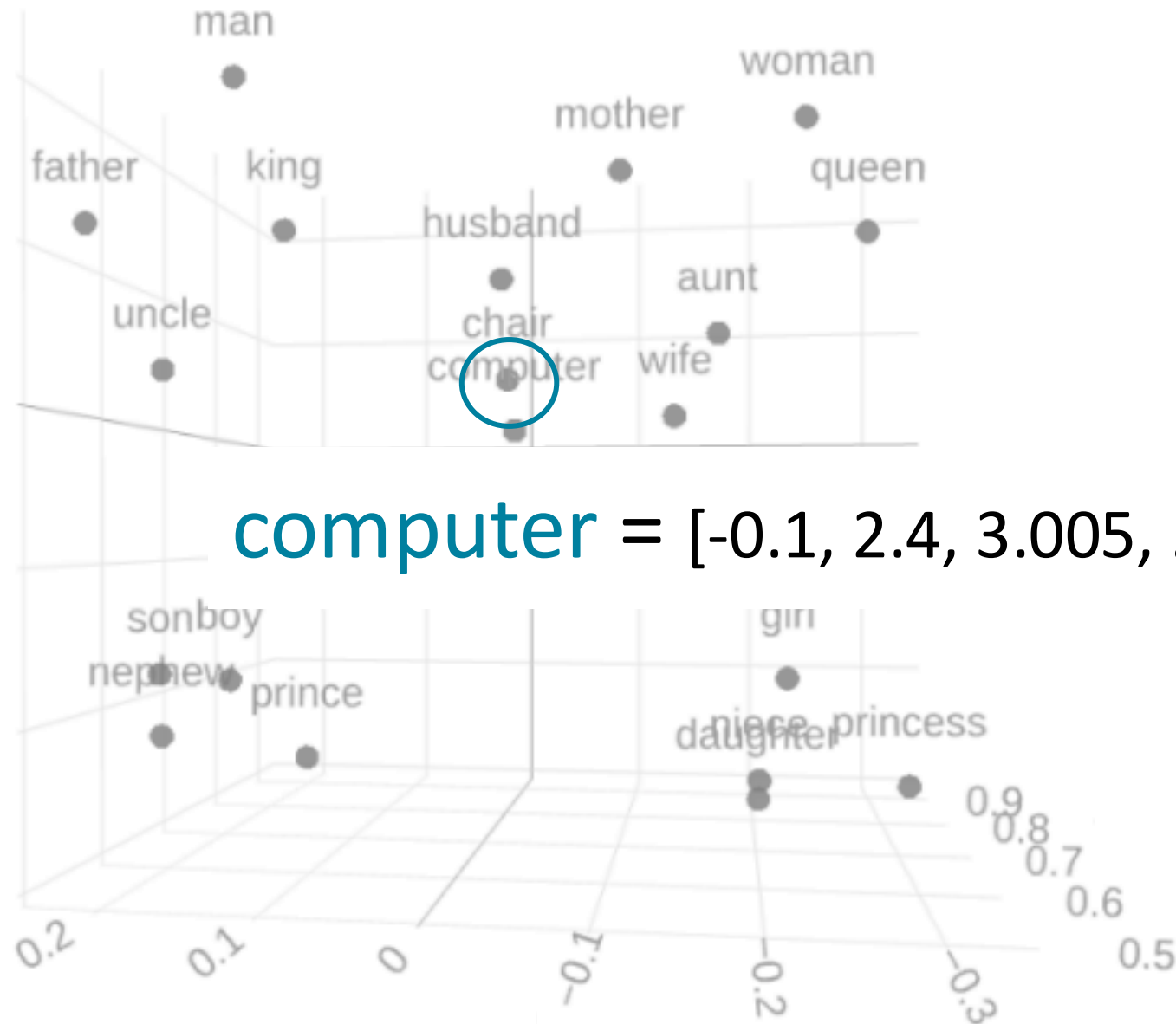
170,000

Aardvark Zyzzyva

1 170,000
Aardvark Zyzzzyva







computer = [-0.1, 2.4, 3.005, ..., 0.07, 100.04]

How LLMs Learn Word Embeddings

She _____ a pet dog.

How LLMs Learn Word Embeddings

She _____ a pet dog.

- A) owned
- B) coughed
- C) slept
- D) haircut
- E) ...

How LLMs Learn Word Embeddings

She _____ a pet dog.

- A) owned
- B) coughed
- C) slept
- D) haircut
- E) ...

Input

“she”,
“_____”
“a”,
“pet”,
“dog”



Model



Output

How LLMs Learn Word Embeddings

Input

“she”,
“_____”
“a”,
“pet”,
“dog”



Model



Output

“pasta”



How LLMs Learn Word Embeddings

Input

“she”,
“_____”
“a”,
“pet”,
“dog”



Model



Output

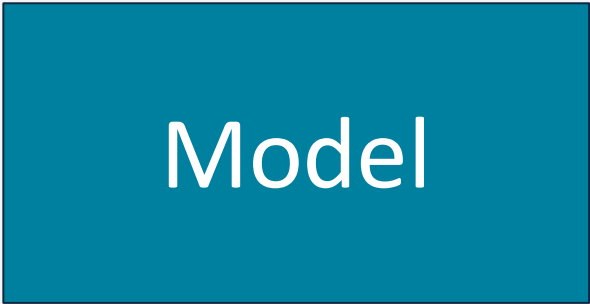
“pasta”



How LLMs Learn Word Embeddings

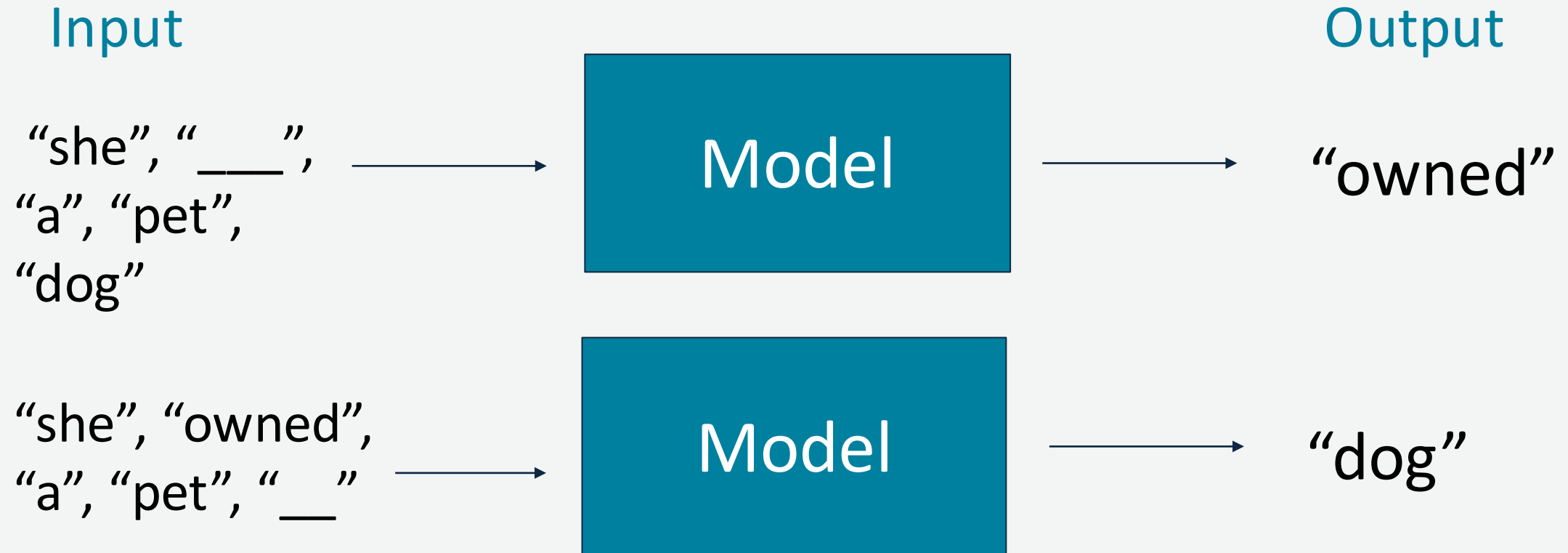
Input

“she”,
“_____”
“a”,
“pet”,
“dog”

Model

Output

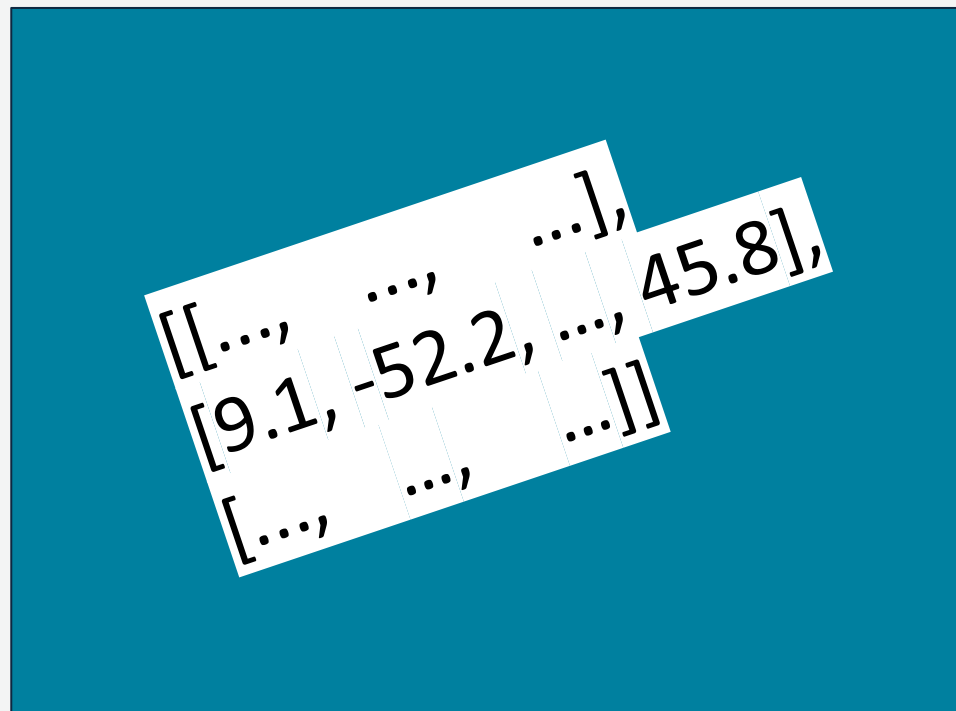
“owned”



How LLMs Learn Word Embeddings

Input

“she”, “a”,
“pet”, “dog”



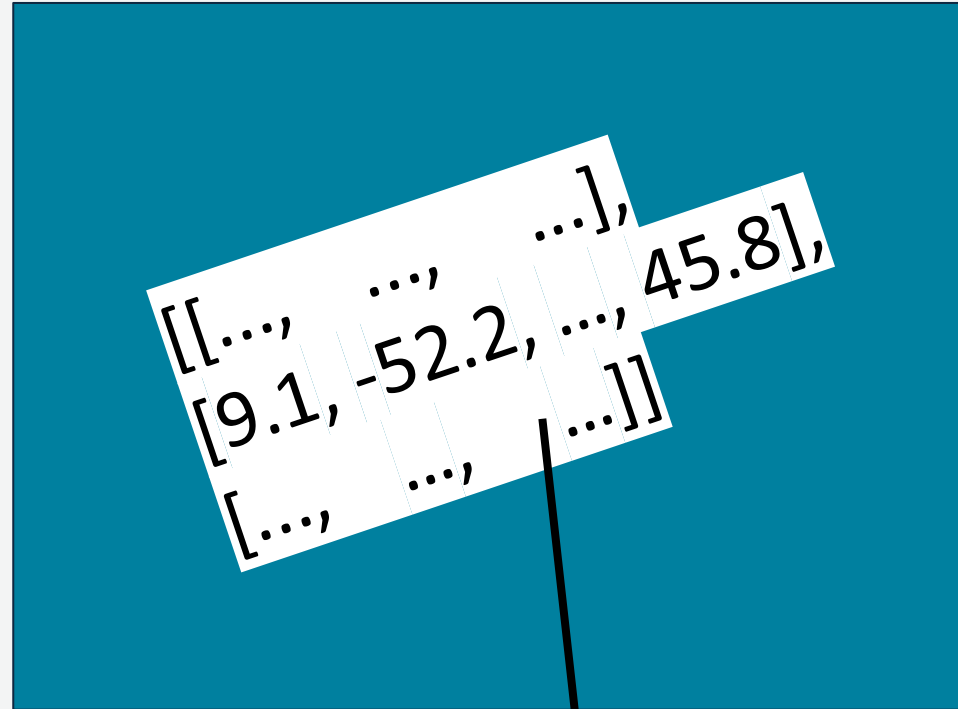
Output

“owned”

How LLMs Learn Word Embeddings

Input

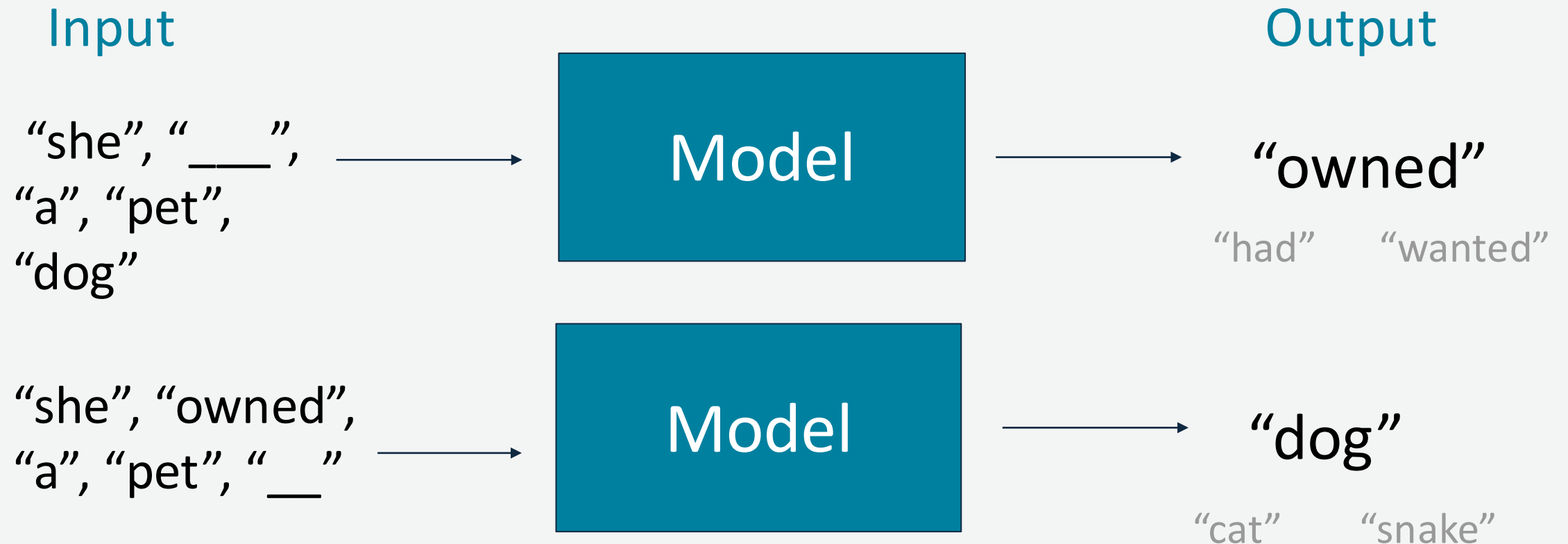
“she”, “a”,
“pet”, “dog”



Output

“owned”

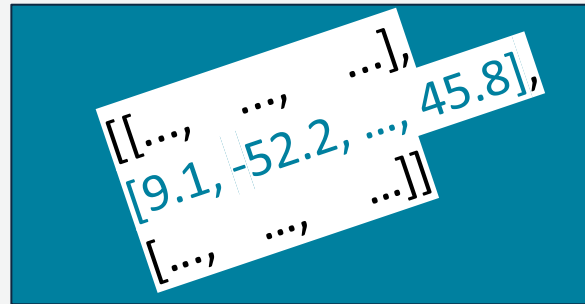
“owned” = $[9.1, -52.2, ..., 45.8]$



How LLMs Learn Word Embeddings

Input

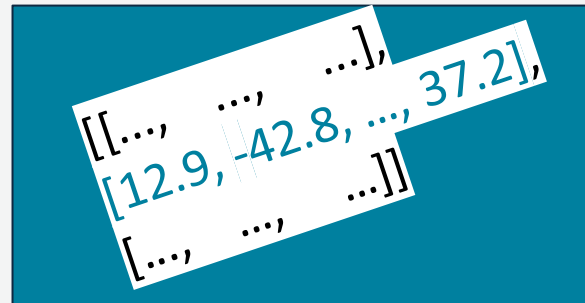
“she”, “____”,
“a”, “pet”,
“dog”



Output

“owned”

“she”, “owned”,
“a”, “pet”, “__”



“had”

How LLMs Learn Word Embeddings

Input

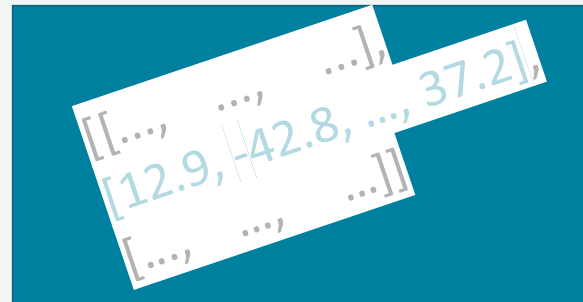
“she”, “____”,
“a”, “pet”,
“dog”



Output

“owned”

“she”, “owned”,
“a”, “pet”, “____”



“had”

One more thing...

Sometimes words aren't the smallest unit of meaning

This is where **tokenization** comes in

Tokenization

- **Simple:** Tokens are just another name for words that are fed into a large language model

GPT-4o & GPT-4o miniGPT-3.5 & GPT-4GPT-3 (Legacy)

Many words map to one token, but some don't.

Words that are made up of smaller parts of meaning might get multiple tokens, like indivisible, soundly, and irrefutable.

ClearShow example

Tokens

38

Characters

167

Many words map to one token, but some don't.

Words that are made up of smaller parts of meaning might get multiple tokens, like indivisible, soundly, and irrefutable.

Tokenization

- **Simple:** Tokens are just another name for words that are fed into a large language model
- **More Nuanced:** Tokenization is how text is split up and fed into large language models. Many words map to one token, but some words are split into multiple tokens.

GPT-4o & GPT-4o miniGPT-3.5 & GPT-4GPT-3 (Legacy)

Many words map to one token, but some don't.

Words that are made up of smaller parts of meaning might get multiple tokens, like indivisible, soundly, and irrefutable.

ClearShow example

Tokens

38

Characters

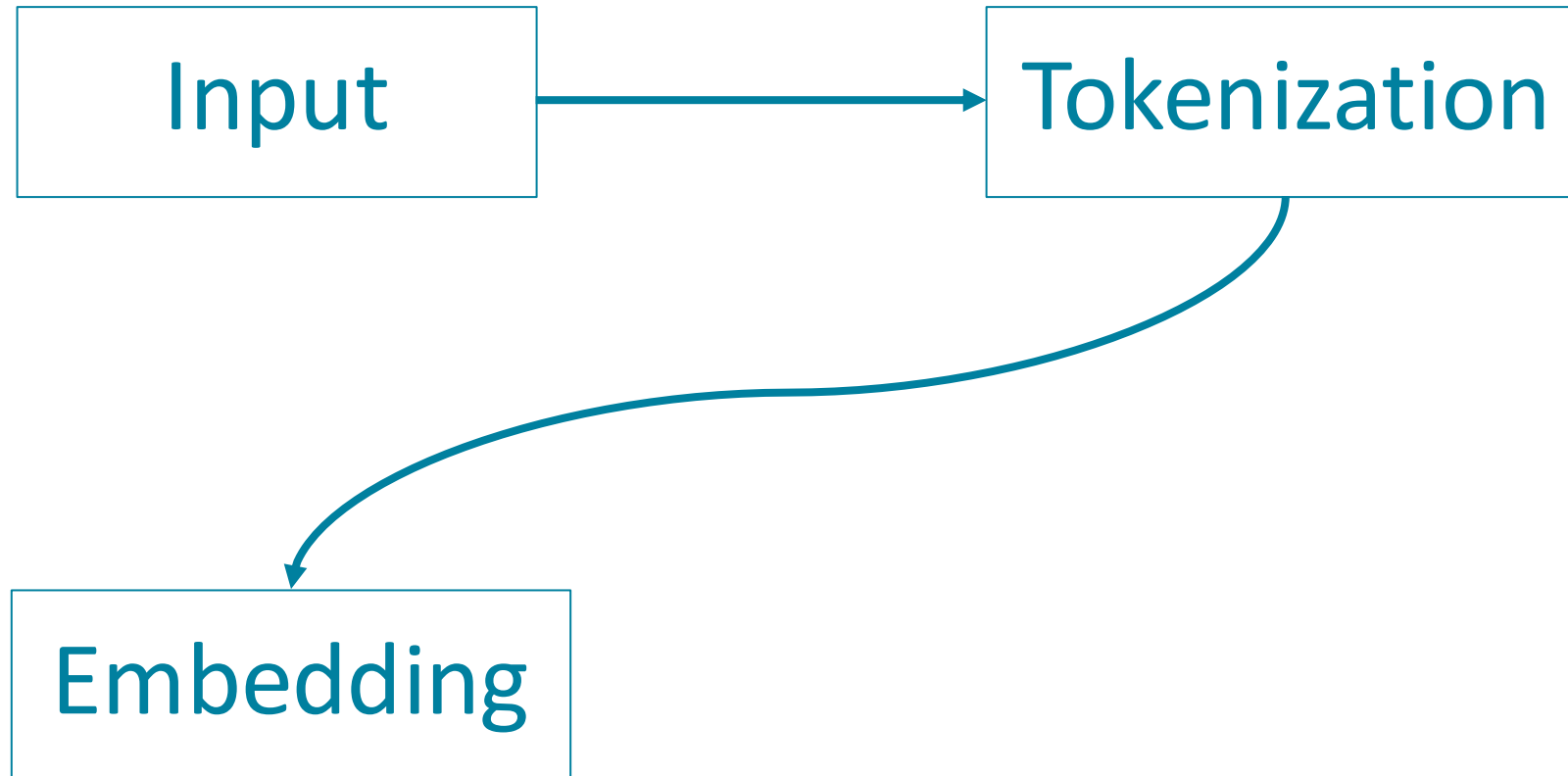
167

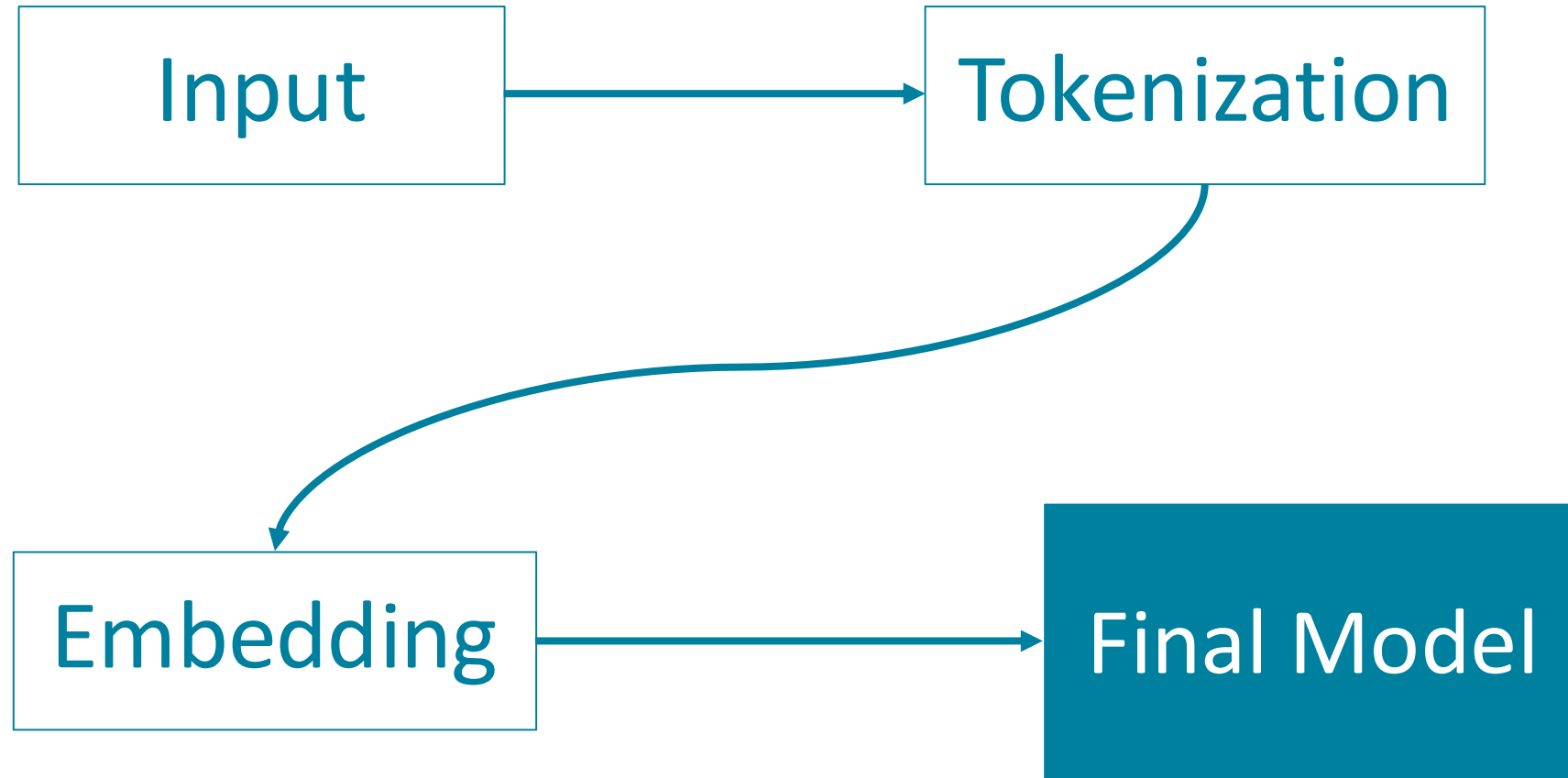
Many words map to one token, but some don't.

Words that are made up of smaller parts of meaning might get multiple tokens, like indivisible, soundly, and irrefutable.

Input







It was the best
of times, it was
the worst of
times...

It was the best
of times, it was
the worst of
times...



“It”, “was”, “the”, “best”,
“of”, “times”, “it”, “was”,
“the”, “worst”, “of”,
“times”

It was the best
of times, it was
the worst of
times...

→ “It”, “was”, “the”, “best”,
“of”, “times”, “it”, “was”,
“the”, “worst”, “of”,
“times”

←
[-0.7,...,126.8],
[-0.8,...,10.7],
...,
[-43.6,...,600.8]

It was the best
of times, it was
the worst of
times...

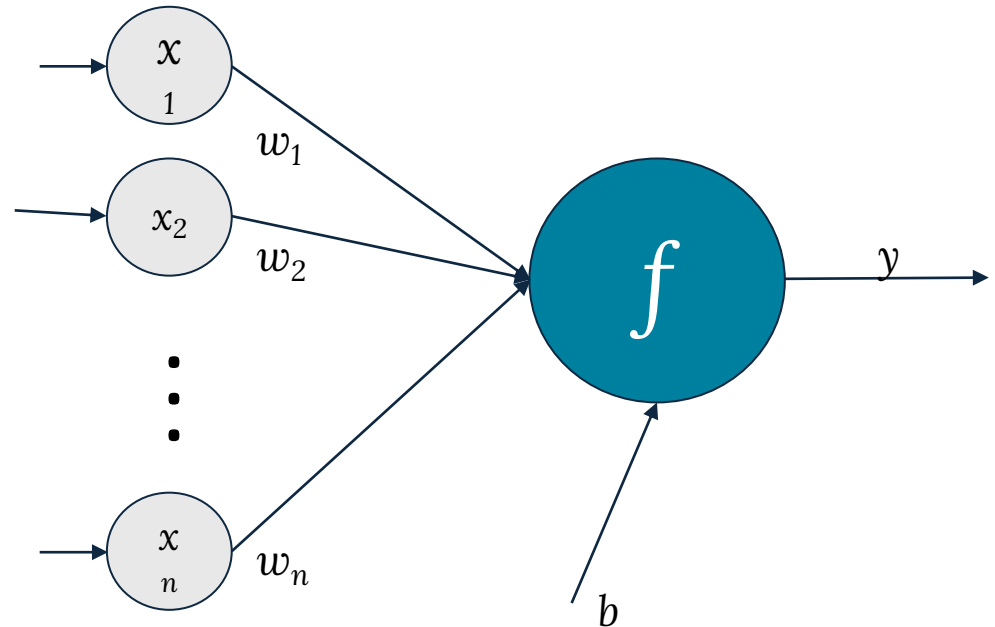
“It”, “was”, “the”, “best”,
“of”, “times”, “it”, “was”,
“the”, “worst”, “of”,
“times”

$[-0.7, \dots, 126.8]$,
 $[-0.8, \dots, 10.7]$,
 $\dots$,
 $[-43.6, \dots, 600.8]$

$[-0.7, \dots, 126.8]$

$[-0.8, \dots, 10.7]$

$[-43.6, \dots, 600.8]$



queen

=

[-2.5, 0.7, ..., 2.8]

queen

=

[-2.5,

0.7,

... ,

2.8]

part of speech?

queen

=

[-2.5,

0.7,

... ,

2.8]

gender?

queen

=

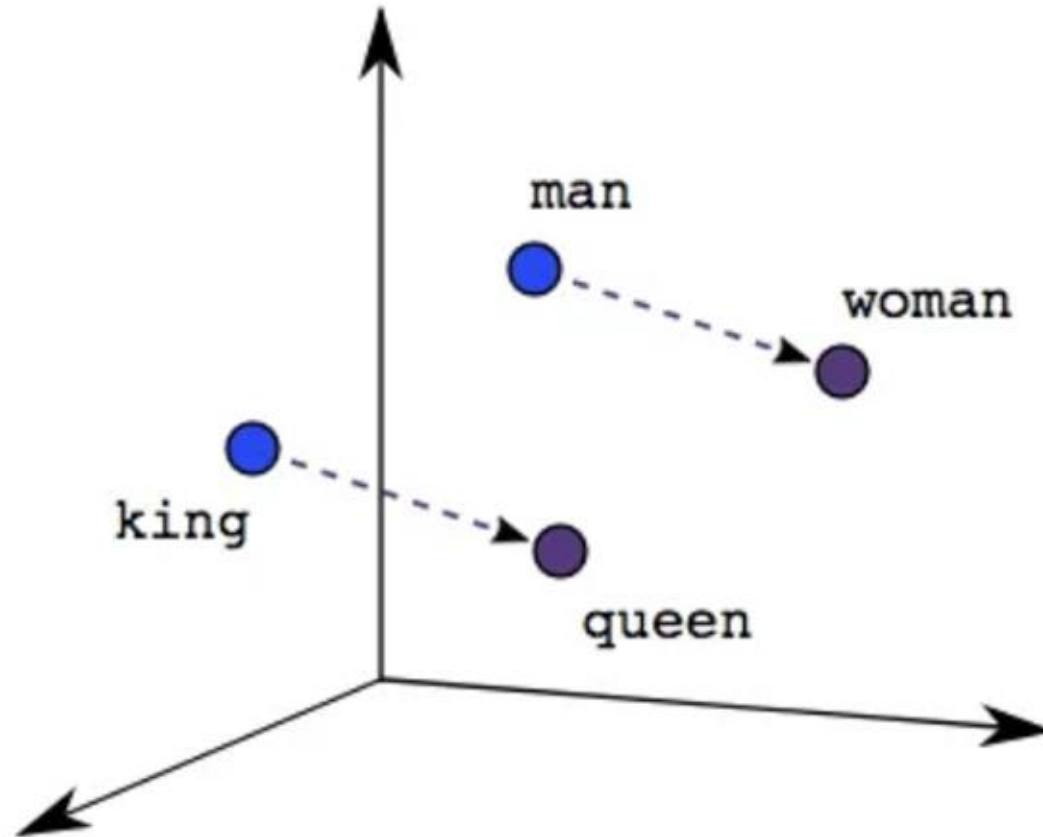
[-2.5,

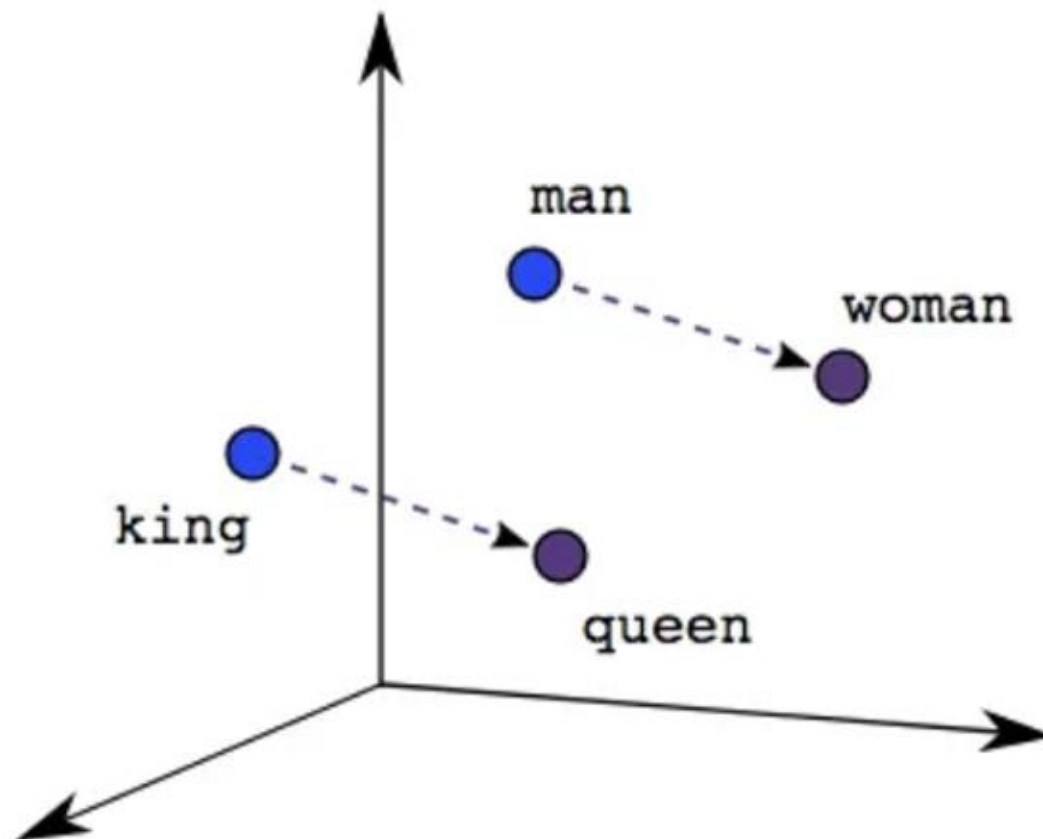
0.7,

... ,

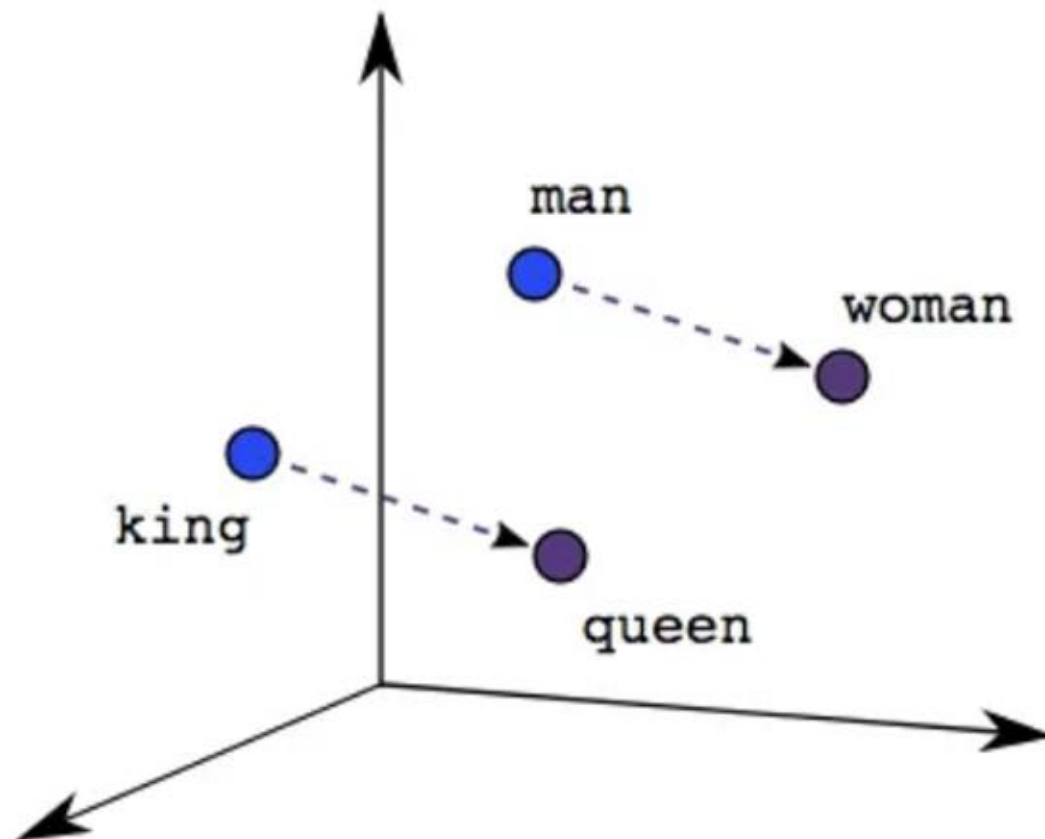
2.8]

occupation?





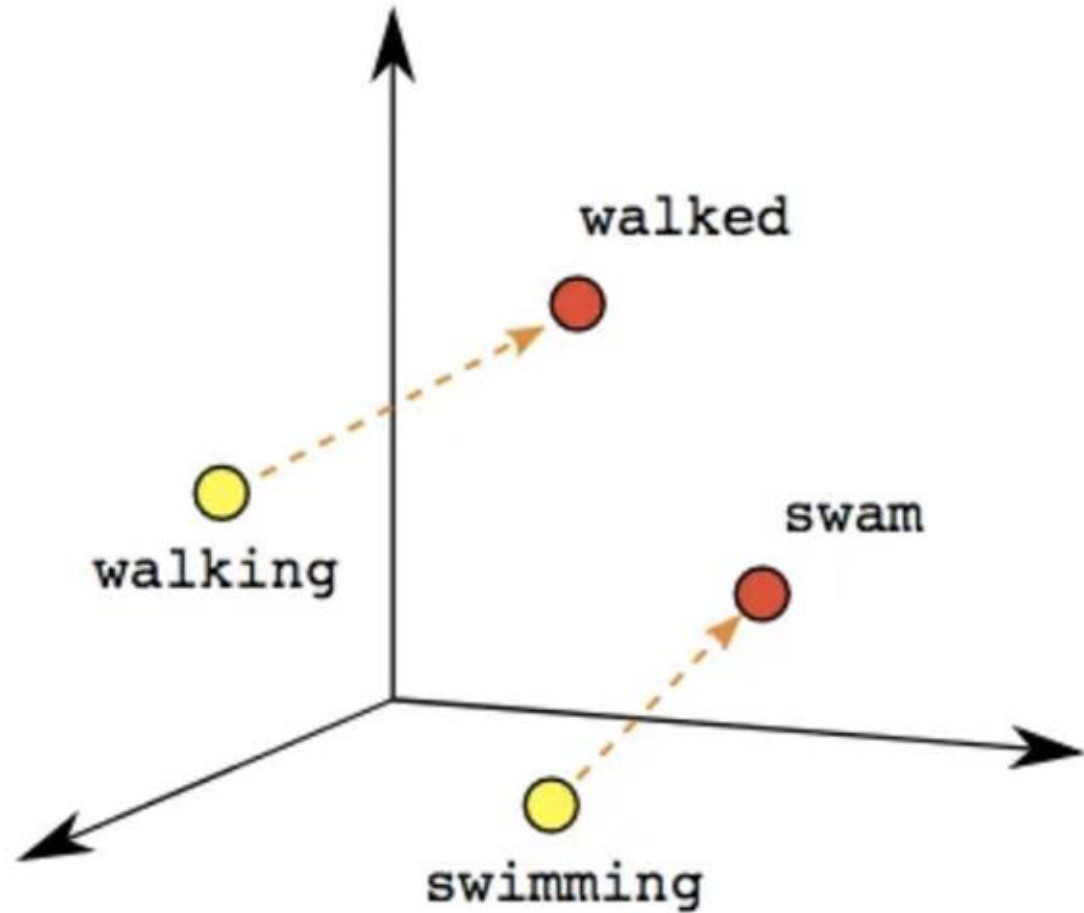
$$\text{king} - \text{man} + \text{woman} = \text{queen}$$



king - man + woman = queen

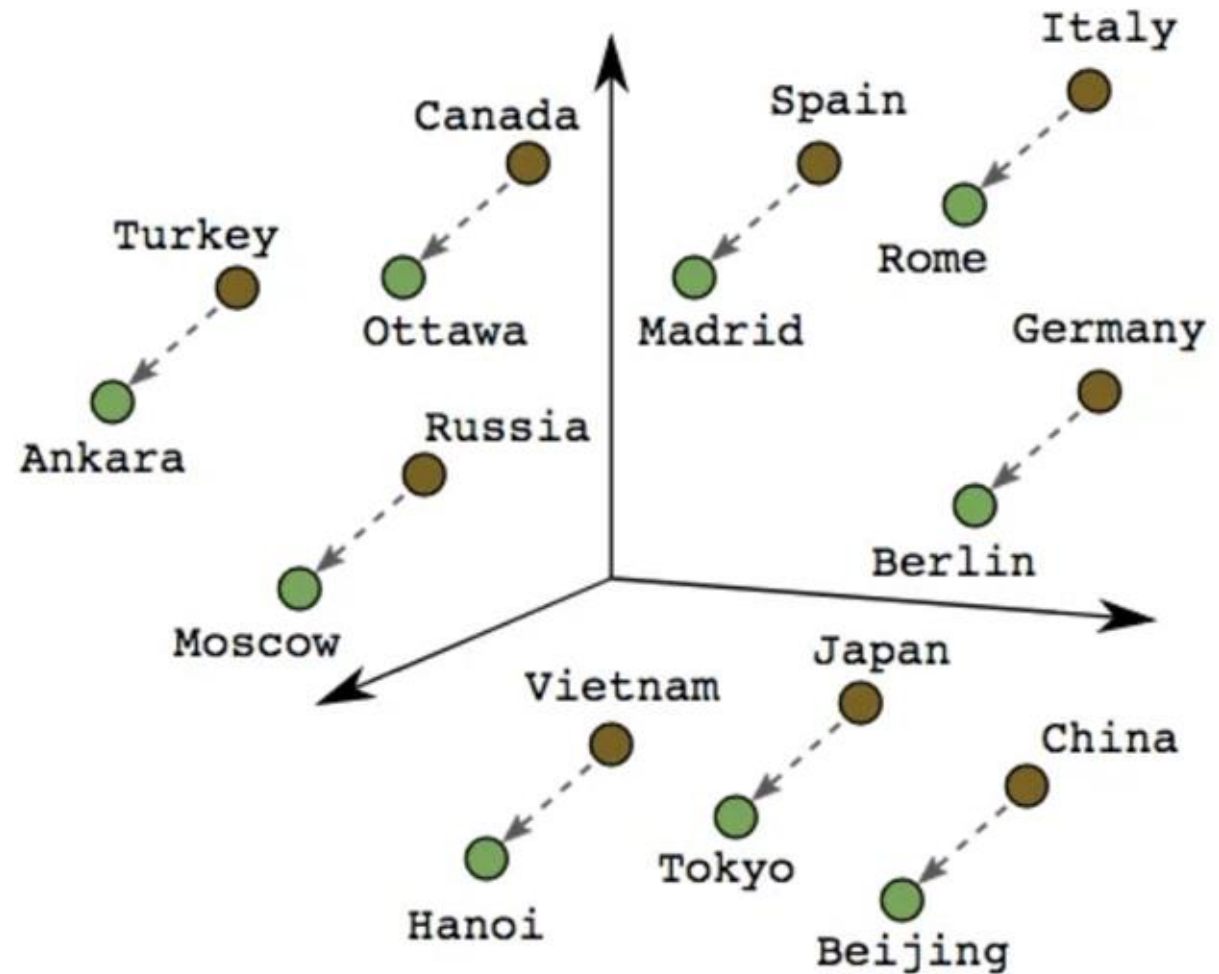
$[-2.5, \dots, 3.1] - [1.5, \dots, 3.1] + [1.5, \dots, 2.8] = [-2.5, \dots, 2.8]$

Many different
analogies can be
turned into **equations**



$$\text{walked} - \text{swam} + \text{swimming} = \text{walking}$$

Many different
analogies can be
turned into **equations**



$$\text{Italy} - \text{Spain} + \text{Madrid} = \text{Rome}$$

Audience Poll Question

LLMs learn word embeddings (multiple choice):

- By seeing lots of textual examples
- Because they are programmed into the model
- By learning which words are likely to appear in similar contexts
- With the help of dictionaries

Audience Poll Question

The LLM pipeline order is (single choice):

- Embedding -> Tokenization -> Language Modeling Task -> Large Language Model
- Tokenization -> Embedding -> Language Modeling Task -> Large Language Model
- Language Modeling Task -> Embedding -> Tokenization -> Large Language Model